MECH230 Dynamics Notes

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Preface

These notes are prepared for the MECH230 Dynamics course at the American University of Beirut (AUB), Lebanon. The content is adapted primarily from O'Reilly's Dynamics Primer, with selected examples and exercises drawn from MKB's and Hibbeler's classic textbooks.

These notes begin with simpler examples to build a strong foundation, enabling students to confidently tackle more complex problems. They are not comprehensive; students should refer to the course textbooks as needed.

I am grateful for the perspectives on dynamics of Professor Noel Perkins (University of Michigan–Ann Arbor), Professor James Casey (UC Berkeley), and Dr. Evan Hemingway, which I gained while TAing Berkeley's ME104.

If you find any mistakes in these notes, please let me know by filling out this form.

Part I

Preliminaries

1 Active Learning

Check out these two excellent videos of Professor Noel Perkins from the University of Michigan at Ann Arbor explaining the active learning techniques he uses to teach dynamics: Part 1, Part 2. We will be using these techniques ourselves. The techniques include:

- **Having an agenda for the day.** Students can add topics they want to discuss too.
- **Frequently checking student understanding:**
  - "Raise your hand if you understand 90% of this or above."
- **Encouragement:**
  - "Ask me a question."
  - "Persevere. Someone else is going to have this question."
  - "I had a great question from one of your classmates."
- **Think → Share:**
  - "I would like you to do the first step on your own. Take a min."
  - "Show of hands: I did it, or I know how to do it."
- **Think → Pair → Share:** * "I would like you to take 5 min to talk to your neighbor and come up with a strategy that will solve this problem."
  - "Show of hands: my neighbor convinced me this will work."
  - "Show of hands: I am not convinced yet."

2 A Note on Units

Quantity	Dimensional Symbol	SI Units		U.S. Customary Units	
		Unit	Symbol	Unit	Symbol
Mass	M	Base units: kilogram	kg	slug	—
Length	L	Base units: meter*	m	Base units: foot	ft
Time	T	Base units: second	s	Base units: second	sec
Force	F	newton	N	Base units: pound	lb

*Also spelled *metre*.

Figure 2.1: Source: MKB

In this class, we will use both SI and US customary units.

In the US customary units, the pound is used both as a unit of force (lbf) and as a unit of mass (lbm). The lbm is the amount of mass which weighs 1 lbf under standard conditions (at latitude 45° and sea level). In MKB, almost exclusively the unit of slug is used for mass. In addition, they use the symbol lb to mean pound force.

Notice that seconds is abbreviated as s in SI units and sec in US customary units.

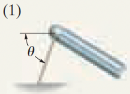
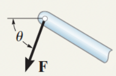
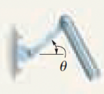
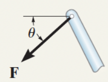
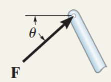
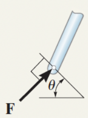
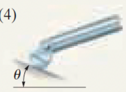
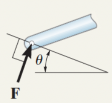
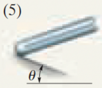
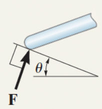
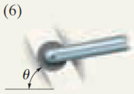
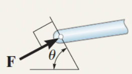
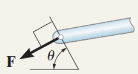
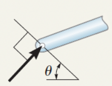
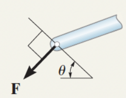
For the gravitational acceleration use the values $g = 9.81$ m s^{-2} in SI units and $g = 32.2$ ft sec^{-2} in US customary units.

> **!** Note!
>
> When solving a problem, keep all variables in the unit system they are given. For example, you may convert g to kg, but don't convert g to slugs!

3 Free Body Diagrams

Recall the following table taken from the Hibbeler Statics textbook. It summarizes the constraint resultant forces and moments due to various types of supports.

TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(1)  cable		One unknown. The reaction is a tension force which acts away from the member in the direction of the cable.
(2)  weightless link	 or 	One unknown. The reaction is a force which acts along the axis of the link.
(3)  roller		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(4)  rocker		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(5)  smooth contacting surface		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(6)  roller or pin in confined smooth slot	 or 	One unknown. The reaction is a force which acts perpendicular to the slot.
(7)  member pin connected to collar on smooth rod	 or 	One unknown. The reaction is a force which acts perpendicular to the rod.

continued

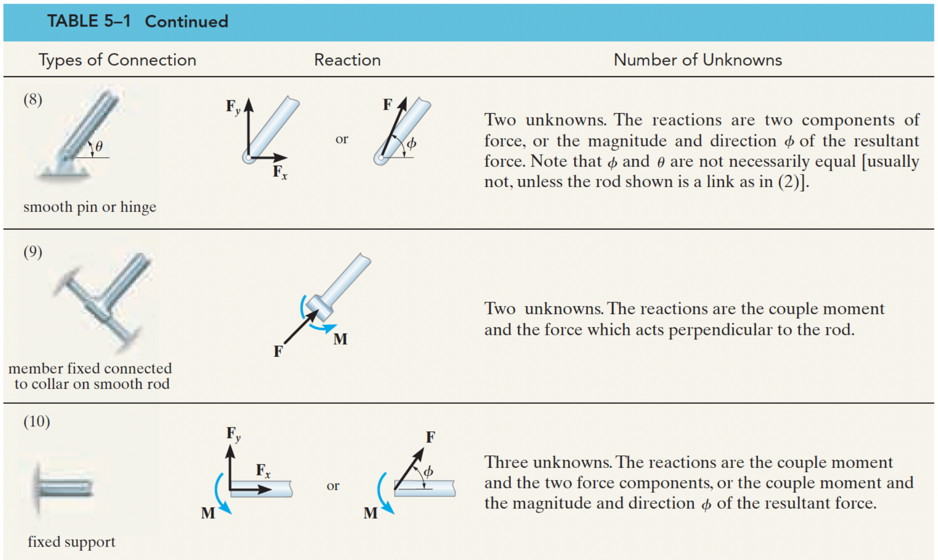


TABLE 5–1 Continued		
Types of Connection	Reaction	Number of Unknowns
(8) smooth pin or hinge	$\mathbf{F}_y$, $\mathbf{F}_x$ or $\mathbf{F}$, ϕ	Two unknowns. The reactions are two components of force, or the magnitude and direction ϕ of the resultant force. Note that ϕ and θ are not necessarily equal [usually not, unless the rod shown is a link as in (2)].
(9) member fixed connected to collar on smooth rod	$\mathbf{F}$, $\mathbf{M}$	Two unknowns. The reactions are the couple moment and the force which acts perpendicular to the rod.
(10) fixed support	$\mathbf{F}_y$, $\mathbf{F}_x$, $\mathbf{M}$ or $\mathbf{F}$, ϕ, $\mathbf{M}$	Three unknowns. The reactions are the couple moment and the two force components, or the couple moment and the magnitude and direction ϕ of the resultant force.

Think!

Question: What is the general rule which encompasses all the cases in the above table and more?

Answer

Whenever translation in a certain direction is constrained, a constraint force in that direction enforces the constraint. Whenever rotation along a certain axis is constrained, a constraint moment along the same axis results.

Note! Notation

Good notation is important: in print, vectors are usually denoted by boldface, e.g. $\mathbf{v}$ or $\mathbf{F}$. In writing, vectors are either denoted by an overhead arrow, e.g. $\vec{v}$, or by a line or squiggle underneath, e.g. $\underset{\sim}{v}$. We opt for the latter notation since we reserve the top of a vector for other accents.

4 Vector Calculus

4.1 Scaling Vectors

We can scale the vector $\mathbf{a}$.

Think!

Question: Consider a vector $\mathbf{a}$. Plot the vectors $2\mathbf{a}$, $-\mathbf{a}$, $\frac{1}{2}\mathbf{a}$

Answer

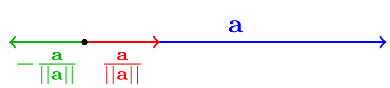
...

Think!

Question: Give the expression for a unit vector in the direction of $\mathbf{a}$? Opposite to $\mathbf{a}$.

Answer

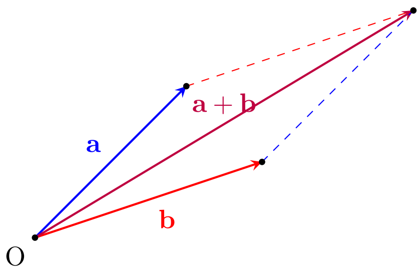
- $\frac{\mathbf{a}}{\|\mathbf{a}\|}$ is a unit vector in the direction of $\mathbf{a}$.
- $-\frac{\mathbf{a}}{\|\mathbf{a}\|}$ is a unit vector in the direction opposite to that of $\mathbf{a}$.



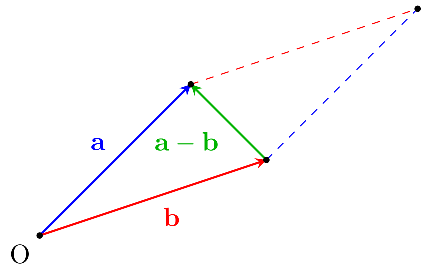
4.2 Vector summation and subtraction

Vectors can be added using the parallelogram law or by placing vectors in a head to tail sequence.

The vector difference $\mathbf{a} - \mathbf{b}$ points from the tip of $\mathbf{b}$ to the tip of $\mathbf{a}$.



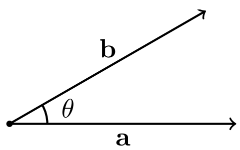
(a) Vector addition via parallelogram law and tip to tail.



(b) Vector subtraction as $\mathbf{a} - \mathbf{b}$ using the parallelogram law. This can be also done using head to tail.

4.3 The dot (scalar) product

Consider the three dimensional Euclidean space $\mathbb{E}^3 = (\mathbb{R}^3, \cdot)$ which is the three-dimensional space $\mathbb{R}^3$ and a dot product defined as follows.



Let $\mathbf{a}, \mathbf{b} \in \mathbb{E}^3$.

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos(\theta). \tag{4.1}$$

Think!

Question: What is the result of $\mathbf{a} \cdot \mathbf{a}$?

Answer

The magnitude of a vector $\mathbf{a}$ is given by

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \implies \|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}. \tag{4.2}$$

Think!

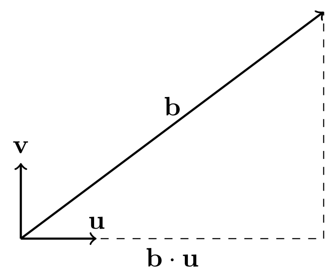
Question: Is $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$? Why?

Answer

Yes, the dot product is commutative. This can be shown using the definition of the dot product.

4.4 Vector Projections

Suppose $\mathbf{u}$ is a unit vector: $\mathbf{u} \cdot \mathbf{u} = 1$.



Think!

Question: Calculate $\mathbf{b} \cdot \mathbf{u}$

Answer

$$\mathbf{b} \cdot \mathbf{u} = \|\mathbf{b}\| \underbrace{\|\mathbf{u}\|}_{1} \cos(\theta) = \|\mathbf{b}\| \cos(\theta). \tag{4.3}$$

Notice the right triangle in the above figure. Verify that $\mathbf{b} \cdot \mathbf{u}$ is the orthogonal projection of $\mathbf{b}$ in the direction of $\mathbf{u}$.

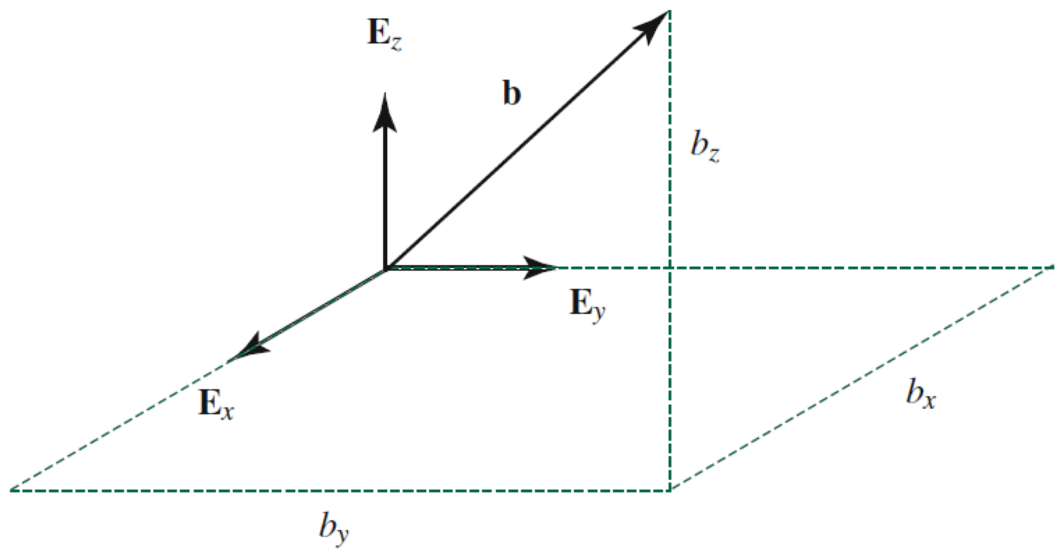
Suppose $\mathbf{v} \cdot \mathbf{v} = 1$ ($\mathbf{v}$ is a unit vector), and $\mathbf{v} \cdot \mathbf{u} = 0$ ($\mathbf{v}$ and $\mathbf{u}$ are orthogonal/perpendicular).

$$\mathbf{b} \cdot \mathbf{v} = \|\mathbf{b}\| \cos\left(\frac{\pi}{2} - \theta\right) = \|\mathbf{b}\| \sin(\theta). \tag{4.4}$$

This is the projection of $\mathbf{b}$ in the direction perpendicular to $\mathbf{u}$. Hence, we can write

$$\begin{aligned} \mathbf{b} &= (\mathbf{b} \cdot \mathbf{u})\,\mathbf{u} + (\mathbf{b} \cdot \mathbf{v})\,\mathbf{v} && (4.5) \\ &= \underbrace{\|\mathbf{b}\|}_{\text{magnitude}} \left(\underbrace{\cos(\theta)\mathbf{u} + \sin(\theta)\mathbf{v}}_{\text{direction}} \right). && (4.6) \end{aligned}$$

Remember that $\cos^2(\theta) + \sin^2(\theta) = 1$.



4.5 Cross Product

We also define the cross product

$$\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \, |\sin(\theta)| \mathbf{n} \tag{4.7}$$

where $\mathbf{n}$ is a unit vector normal to the plane formed by $\mathbf{a}$ and $\mathbf{b}$. The direction of $\mathbf{n}$ is given by the right hand rule.

Think!

Question: Is $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}$?

Answer

No, the cross product is not commutative. These vectors are opposite.

4.6 Vector Representation Using Bases

We can assign a fixed (right handed) Cartesian basis for $\mathbb{E}^3$ denoted by $\{\mathbf{E}_x, \mathbf{E}_y, \mathbf{E}_z\}$. These three vectors are orthonormal (i.e. they each have a unit magnitude and are mutually perpendicular).

Think!

Question: Calculate $\mathbf{E}_x \times \mathbf{E}_x$, $\mathbf{E}_x \times \mathbf{E}_y$, etc.

Answer

...

Think!

Question: Calculate $\mathbf{E}_x \cdot \mathbf{E}_x$, $\mathbf{E}_x \cdot \mathbf{E}_y$, etc.

Answer

...

We can represent any vector $\mathbf{b}$ as

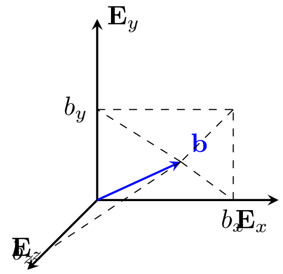
$$\mathbf{a} = a_x\mathbf{E}_x + a_y\mathbf{E}_y + a_z\mathbf{E}_z, \tag{4.8}$$

$$\mathbf{b} = b_x\mathbf{E}_x + b_y\mathbf{E}_y + b_z\mathbf{E}_z. \tag{4.9}$$

We can write the scalar product and cross product of these vectors in terms of their Cartesian components.

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{4.10}$$

$$\mathbf{a} \times \mathbf{b} = \ldots \tag{4.11}$$



4.7 Differentiation of Vectors

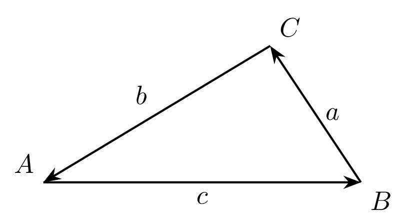
$$\frac{d\mathbf{u}}{dt} = \frac{du_x}{dt}\mathbf{E}_x + \frac{du_y}{dt}\mathbf{E}_y + \frac{du_z}{dt}\mathbf{E}_z + u_x\frac{d\mathbf{E}_x}{dt} + u_y\frac{d\mathbf{E}_y}{dt} + u_z\frac{d\mathbf{E}_z}{dt} \tag{4.12}$$

$$= \frac{du_x}{dt}\mathbf{E}_x + \frac{du_y}{dt}\mathbf{E}_y + \frac{du_z}{dt}\mathbf{E}_z \tag{4.13}$$

$$\frac{d}{dt}(\mathbf{u}\cdot\mathbf{v}) = \frac{d\mathbf{u}}{dt}\cdot\mathbf{v} + \mathbf{u}\cdot\frac{d\mathbf{v}}{dt} \tag{4.14}$$

$$\frac{d}{dt}(\mathbf{u}\times\mathbf{v}) = \frac{d\mathbf{u}}{dt}\times\mathbf{v} + \mathbf{u}\times\frac{d\mathbf{v}}{dt} \tag{4.15}$$

4.8 Sine Law and Cosine Law



Referring to the above triangle with interior angles A, B, and C and with side lengths a, b, and c, the sine law and cosine law are respectively

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}, \tag{4.16}$$

$$c^2 = a^2 + b^2 - 2ab\cos(C). \tag{4.17}$$

5 Fundamental Concepts of Solid Mechanics

This section is adapted from chapter 1 of Introduction to Solid Mechanics - An Integrated Approach by Lubliner and Papadopoulos (Lubliner and Papadopoulos 2017).

5.1 Mechanics

Mechanics, as a scientific discipline, is the study of the behavior of bodies subject to forces and displacements (and, to some extent, heating and cooling).

5.2 Solid Mechanics

Solid mechanics is the branch of mechanics dealing specifically with solid bodies.

5.3 Body

A body, for the purpose of mechanics, is a portion of matter that, at a given moment in time, occupies a certain region in space.

- **Particle** If the region occupied by the body can be idealized as being of negligible extent (and thus reducible to a point), the body is called a particle.
- **Finite Body** A body that is not reducible to a point is called a finite body.

5.4 Continuum

A finite body occupying a connected region every portion of which contains some matter is called a continuous body or, simply, a continuum.

5.5 Material Points

At any point in a region occupied by a continuum, the matter in its immediate (infinitesimal) neighborhood can also be thought of as constituting a particle, and we may thus speak of the particles (also called material points) of a continuum.

5.6 Displacement

A body is said to undergo a displacement when some or all of its particles are moved to occupy different positions in space.

5.7 Configuration

The correspondence between the particles and the positions occupied by them at a given time is known as the body's configuration.

5.8 Deformation

The displacement of a body is said to be rigid if the distances between all pairs of particles are the same in any two configurations. Otherwise, the body is said to undergo deformation.

5.9 Rigid bodies

Bodies that can undergo only rigid displacements are called rigid bodies. Bodies that are not rigid are deformable.

5.10 Motion

Motion is said to occur when the body occupies a succession of different configurations continuously over time.

5.11 Rigid Motion

A rigid motion is a motion that involves only rigid displacements.

5.12 Velocity

The rate of change, in time, of the position of a particle is its velocity.

5.13 Acceleration

The velocity may in turn change in time, and its rate of change is the acceleration. The motion of a body is called uniform if the acceleration of all the particles is zero; otherwise, it is called accelerated.

5.14 Kinematics

The branch of mechanics dealing with the displacement and motion of bodies is called kinematics.

5.15 Force

Forces are interactions between bodies that cause them to move, and more specifically to accelerate, relative to each other (unless prevented from doing so by other forces).

6 Galileo, Kepler, Newton

6.1 Gravitational force

All bodies on or near the earth are subject to the earth's gravitational force and, as shown by Galileo in his famous experiments, will fall toward the center of the earth (unless prevented from doing so by other forces) with the same acceleration; the magnitude of this acceleration is denoted by g.

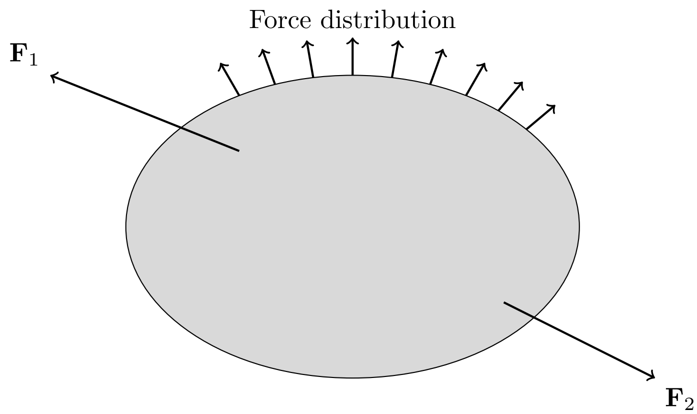
It was later shown by Newton, on the basis of Kepler's laws of planetary motion, that the magnitude of the acceleration (with respect to a frame of reference based on the "fixed" stars) due to gravitational force between any two bodies (idealized as particles) is inversely proportional to the square of the distance between them, and for each body it is proportional to the quantity of matter (or mass) of the other body. The product of mass and acceleration is therefore equal and opposite for the two bodies and may be identified with the force exerted on each body by the other.

6.2 Equilibrium

If a body's acceleration is zero, then it is at rest or in a uniform motion. In this case, the forces is said to be in equilibrium. The study of equilibrium is known as statics and constitutes one of the main subjects of this book. Traditionally, statics deals only (or at least primarily) with rigid bodies, while deformable bodies are studied in courses historically called strength of materials (or, more recently, mechanics of materials). In this book (Lubliner and Papadopoulos 2017), the statics of rigid and deformable solid bodies will be studied in an integrated manner.

7 Starting from Continuum Mechanics

Consider a body $\mathscr{B}$ in three-dimensional Euclidean space $\mathbb{E}^3$.

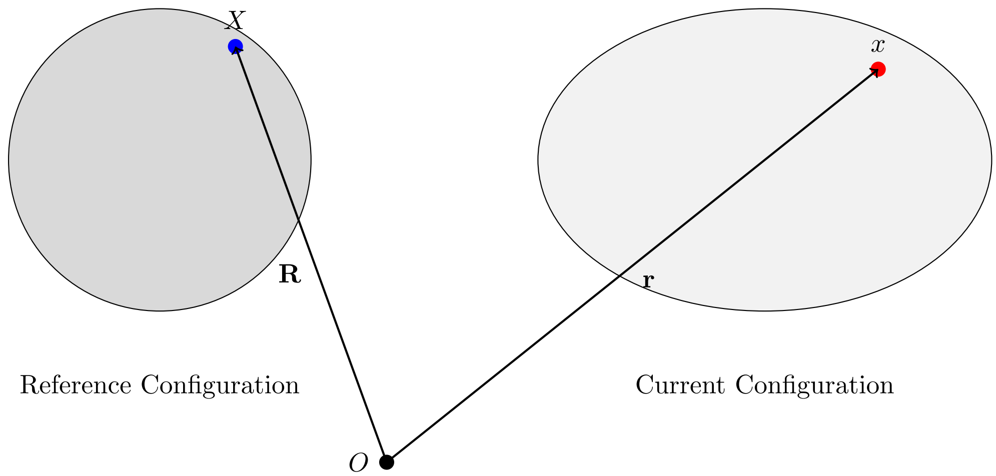


In general, bodies are deformable continua. We depict continua as in the above figure and refer to them as *continuum potatoes*.

Point or distributed forces or moments can be applied to a body's surface or throughout its volume. For example, a body resting on a table has contact forces on the surface touching the table, while weight acts on all points in the body.

> Think!
>
> **Question:** How do we know if a body has moved?
>
> **Answer**
>
> ...

Consider a reference configuration for this body. A reference configuration is any configuration that can be physically occupied by the body — it could be the initial configuration, but need not be.



We denote the position vector from a chosen fixed origin to a material point in the reference configuration as $\mathbf{R}$. The position vector to the same material point in the current configuration is $\mathbf{r}$.

The motion of the body is described by

$$\mathbf{r} = \chi(\mathbf{R}, t). \tag{7.1}$$

Think!

Question: What makes a body rigid?

Answer

In this class, we consider only rigid bodies. If we choose two points on a rigid body, the distance between them is always constant. Also, angles between material lines on a rigid body are constant.

Think!

Question: Under what conditions do we consider a rigid body to be a particle?

Answer

In the beginning of the class, we consider particles. Particles are idealized rigid bodies whose rotational inertia is ignored.

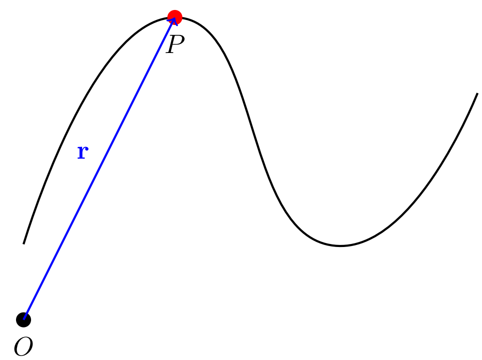
Part II

Particle Dynamics

8 Basic Particle Kinematics Concepts

8.1 Position, Velocity, and Acceleration

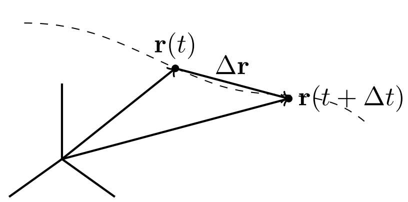
Consider a particle moving in $\mathbb{E}^3$. Its position vector relative to a fixed origin O is denoted by $\mathbf{r} \in \mathbb{E}^3$. As time t evolves, so does $\mathbf{r}(t)$.



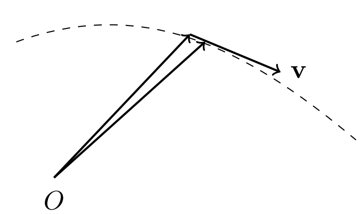
Let Δt be a finite time elapsed:

$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t). \tag{8.1}$$

As $\Delta t \to 0$, Δt becomes dt and $\Delta \mathbf{r}$ becomes $d\mathbf{r}$.



zoom in



The (absolute) velocity vector $\mathbf{v}$ of the particle is

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}. \tag{8.2}$$

Think!

Question: What is the direction of the velocity vector?

Answer

$\mathbf{v}$ is always tangent to the path.

The speed of the particle is the magnitude of the velocity vector $v = \|\mathbf{v}\|$.

The (absolute) acceleration is

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}} = \ddot{\mathbf{r}} = \frac{d^2\mathbf{r}}{dt^2}. \tag{8.3}$$

8.2 Arc-Length and the Unit Tangent Vector

Define the arc-length parameter s such that

$$\frac{ds}{dt} = \|\mathbf{v}\|. \tag{8.4}$$

$\frac{ds}{dt}$ is the speed of the particle. Recall that $\mathbf{v} = \frac{d\mathbf{r}}{dt}$, so

$$\frac{ds}{dt} = \left\|\frac{d\mathbf{r}}{dt}\right\|. \tag{8.5}$$

So ds is the length of the differential $d\mathbf{r}$ along the curve. Integrating:

$$s(t) - s_0 = \int_{t_0}^{t} \frac{ds}{dt}(\tau) d\tau = \int_{t_0}^{t} \sqrt{\mathbf{v}(\tau)\cdot\mathbf{v}(\tau)}\, d\tau. \tag{8.6}$$

Notes:

- $s(t) - s_0$ is the distance traveled by the particle along its path $\mathcal{C}$ during the time interval $[t_0, t]$.
- $s(t_0) = s_0$ where t_0 and s_0 are initial conditions.
- The dummy variable τ is used (as opposed to t) when evaluating this integral because we are integrating the speed as it varies between t_0 and t.

Think!

Question: Under what conditions is $s(t)$ invertible so that $t(s)$ is well defined?

Answer

If the function $s(t)$ is one-to-one, it can be inverted, and time can be written in terms of arc-length $t = t(s)$.

Think!

Question: Why is

$$s(t) - s_0 \neq \|\mathbf{r}(t) - \mathbf{r}_0\|\,? \tag{8.7}$$

Under what conditions does the equality hold?

Answer

The equality holds under rectilinear motion or over an infinitesimal time interval.

We define the unit tangent vector $\mathbf{e}_t$:

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} = \mathbf{e}_t, \qquad \mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \dot{s}\mathbf{e}_t. \tag{8.8}$$

Think!

Question: Verify that $\mathbf{e}_t$ is a unit vector.

Answer

...

Note the relationships

$$\begin{aligned} \mathbf{v} &= \frac{d\mathbf{r}(t)}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt}, \\ \mathbf{a} &= \frac{d\mathbf{v}(t)}{dt} = \frac{d^2\mathbf{r}}{ds^2}\left(\frac{ds}{dt}\right)^2 + \frac{d\mathbf{r}}{ds}\frac{d^2 s}{dt^2}. \end{aligned}$$

Notice: the acceleration vector is due to changes in both the magnitude and direction of the velocity vector.

Think!

Question: Is the unit vector $\mathbf{e}_t$ constant in general? Under what conditions is it constant?

Answer

No. It is constant only during rectilinear motion in a fixed direction.

8.2.1 Example

Consider a particle moving in space with position vector

$$\mathbf{r}(t) = R_0\left(\cos(\omega t)\mathbf{E}_x + \sin(\omega t)\mathbf{E}_y\right). \tag{8.9}$$

Calculate and plot $\mathbf{v}$, $\mathbf{a}$, $\mathbf{v} \times \mathbf{a}$, $\|\mathbf{r}\|$. Show that the motion satisfies:

$$\ddot{x} + \omega^2 x = 0, \qquad \ddot{y} + \omega^2 y = 0. \tag{8.10}$$

⚠ Example

Question: Consider a particle P moving in $\mathbb{E}^2$ with position vector

$$\mathbf{r} = R_0\left(\cos(\omega t)\,\mathbf{E}_x + \sin(\omega t)\,\mathbf{E}_y\right).$$

Calculate $\mathbf{r} \times \mathbf{a}$ and $\mathbf{r} \cdot \mathbf{a}$. Describe the motion of the particle.

Answer

The velocity and acceleration are:

$$\begin{aligned}\mathbf{v} &= R_0\omega\left(-\sin(\omega t)\,\mathbf{E}_x + \cos(\omega t)\,\mathbf{E}_y\right),\\ \mathbf{a} &= -R_0\omega^2\left(\cos(\omega t)\,\mathbf{E}_x + \sin(\omega t)\,\mathbf{E}_y\right).\end{aligned}$$

Note that $\dot{\mathbf{E}}_x = \mathbf{0}$ and $\dot{\mathbf{E}}_y = \mathbf{0}$ since the basis is fixed, and the chain rule applies.

Noting that $\mathbf{a} = -\omega^2\mathbf{r}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times (-\omega^2\mathbf{r}) = -\omega^2\mathbf{r} \times \mathbf{r} = \mathbf{0}.$$

For $\mathbf{r} \cdot \mathbf{a}$:

$$\mathbf{r} \cdot \mathbf{a} = -R_0^2\omega^2\left(\cos^2(\omega t) + \sin^2(\omega t)\right) = -R_0^2\omega^2.$$

8.3 Cartesian Coordinates

$\{\mathbf{E}_x, \mathbf{E}_y, \mathbf{E}_z\}$ is a basis for $\mathbb{E}^3$. Since this basis is fixed, we have:

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z, \tag{8.11}$$
$$\mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y + \dot{z}\mathbf{E}_z, \tag{8.12}$$
$$\mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y + \ddot{z}\mathbf{E}_z. \tag{8.13}$$

Later we will also use the cylindrical polar basis $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{E}_z\}$ and the Serret-Frenet triad $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

8.4 Rectilinear Motion

- *rectus* = straight, *linea* = line

Due to constraints or initial conditions, the body moves on a straight line, e.g. a car on a straight road or a ball tossed vertically.

We take $\mathbf{E}_x$ parallel to the line and $\mathbf{c}$ to be a constant vector. Then:

$$\mathbf{r}(t) = x(t)\mathbf{E}_x + \mathbf{c}, \tag{8.14}$$

$$\mathbf{v}(t) = \dot{x}\mathbf{E}_x = v(t)\mathbf{E}_x, \tag{8.15}$$

$$\mathbf{a}(t) = \ddot{x}\mathbf{E}_x = a(t)\mathbf{E}_x. \tag{8.16}$$

$\frac{ds}{dt} = \left|\frac{dx}{dt}\right|$, so unless $\dot{x} > 0$ or $\dot{x} < 0$ throughout, x and s cannot be easily interchanged.

8.4.1 Identities for Rectilinear Motion

Recall $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$. Applying the chain rule:

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}. \tag{8.17}$$

Three useful identities:

$$ds = v\,dt, \qquad dv = a\,dt, \qquad v\,dv = a\,ds. \tag{8.18}$$

8.4.2 Given Acceleration as a Function of Time

From $dv = a\,dt$:

$$v(t) - v(t_0) = \int_{t_0}^{t} a(\tilde{t})\,d\tilde{t}. \tag{8.19}$$

From $ds = v\,dt$:

$$s(t_1) - s(t_0) = \int_{t_0}^{t_1} v(\tilde{t})\,d\tilde{t}. \tag{8.20}$$

8.4.3 Given Acceleration as a Function of Speed

From $dv = a\,dt$ with $a = a(v)$:

$$t(v) - t(v_0) = \int_{v_0}^{v} \frac{d\tilde{v}}{a(\tilde{v})}. \tag{8.21}$$

Using $a = v\frac{dv}{ds}$:

$$s(t) - s_0 = \int_{v_0}^{v} \frac{\tilde{v}\,d\tilde{v}}{a(\tilde{v})}. \tag{8.22}$$

8.4.4 Given Acceleration as a Function of Displacement

Using $v\,dv = a\,ds$ with $a = a(s)$:

$$\frac{1}{2}\left(v^2(s) - v^2(s_0)\right) = \int_{s_0}^{s} a(\tilde{s})\,d\tilde{s}. \tag{8.23}$$

8.4.4.1 Constant Acceleration

Example: Gravity. Throw a ball up; take the hand as origin. Here $a = -g$.

$$v(t) = v_0 - gt, \tag{8.24}$$

$$s(t) = s_0 + v_0 t - \frac{g}{2}t^2. \tag{8.25}$$

Example: Vehicle Braking. Using $v\,dv = a\,ds$:

$$v^2(s) - v^2(s_0) = 2a\,\Delta s. \tag{8.26}$$

9 Particle Kinetics

9.1 The Balance of Linear Momentum

A particle is endowed with an inertial measure m, called its mass.

The linear momentum of a particle is defined to be

$$\mathbf{G} = m\mathbf{v}. \tag{9.1}$$

Taking the time derivative:

$$\dot{\mathbf{G}} = \dot{m}\mathbf{v} + m\dot{\mathbf{v}}. \tag{9.2}$$

Think!

Question: Can you think of systems whose mass is not conserved?

Answer

In this class, we consider cases where mass is conserved, i.e. $\dot{m} = 0$. This is not the case, for example, for a launching rocket that is losing mass. The $\dot{m}\mathbf{v}$ term is useful in control volume analysis.

Then, in our case of interest:

$$\dot{\mathbf{G}} = m\dot{\mathbf{v}} = m\mathbf{a}. \tag{9.3}$$

We postulate the following balance law:

$$\mathbf{F} = \dot{\mathbf{G}}, \tag{9.4}$$

where $\mathbf{F}$ is the resultant external force acting on the particle. This is known as the **Balance of Linear Momentum**, **Newton's Second Law**, or **Euler's First Law**.

Important: $\mathbf{a}$ here is the absolute acceleration vector of the particle.

Think!

Question: Under what conditions is the linear momentum $\mathbf{G}$ conserved?

Answer

An unbalanced net force $\mathbf{F} \neq \mathbf{0}$ results in a change of $\mathbf{G}$. If $\mathbf{F} = \mathbf{0}$, then $\mathbf{G} = \mathbf{c}$ where $\mathbf{c}$ is a constant vector. This is Newton's First Law.

If $\mathbf{c} = \mathbf{0}$, we recover statics.

9.1.1 Representations and Projections

Vectors are represented on a basis. For example:

$$\mathbf{F} = F_x \mathbf{E}_x + F_y \mathbf{E}_y + F_z \mathbf{E}_z, \tag{9.5}$$

$$\mathbf{a} = a_x \mathbf{E}_x + a_y \mathbf{E}_y + a_z \mathbf{E}_z. \tag{9.6}$$

Think!

Question: How many independent scalar equations can we obtain from $\mathbf{F} = m\mathbf{a}$? How do we obtain them?

Answer

We get useful scalar equations by projecting $\mathbf{F} = \dot{\mathbf{G}}$ along unit directions:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies F_x = m a_x. \tag{9.7}$$

9.2 Newton's Third Law

When dealing with the forces of interaction between particles, a particle and a rigid body, or rigid bodies, we also invoke Newton's third law: "For every action there is an equal and opposite reaction." For example, the force exerted by a surface on a particle is equal in magnitude and opposite in direction to the force exerted by the particle on the surface.

10 Four Steps to Problem Solving Using the Balance of Linear Momentum

1. Specify the system you are considering. Pick a frame (choose an origin and a coordinate system), and express $\mathbf{r}$, $\mathbf{v}$ and $\mathbf{a}$ using that coordinate system.
2. Draw a Free Body Diagram (FBD), i.e. model of the forces.
3. Write the balance of linear momentum for the system: $\mathbf{F} = \dot{\mathbf{G}}$.
4. Do the analysis: project $\mathbf{F} = \dot{\mathbf{G}}$ to get useful equations.

10.1 Example: Projectile Motion

Projectile motion is a curvilinear motion in the plane $\mathbb{E}^2$.

1. Choose O at the launch point, $\mathbf{E}_x$ along the ground, $\mathbf{E}_y$ against gravity.

$$\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y, \quad \mathbf{v} = \dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y, \quad \mathbf{a} = \ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y. \tag{10.1}$$

2. Force model:

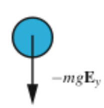


Figure 10.1: FBD

$$\mathbf{W} = mg\left(-\mathbf{E}_y\right). \tag{10.2}$$

3. Balance of linear momentum:

$$mg(-\mathbf{E}_y) = m\left(\ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y\right). \tag{10.3}$$

Projecting:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_x \implies 0 = m\ddot{x} \implies \ddot{x} = 0, \tag{10.4}$$

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right) \cdot \mathbf{E}_y \implies -mg = m\ddot{y} \implies \ddot{y} = -g. \tag{10.5}$$

4. Integrating:

$$\dot{x}(t) = \dot{x}_0, \quad x(t) = \dot{x}_0 t, \tag{10.6}$$

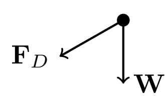
$$\dot{y}(t) = -gt + \dot{y}_0, \quad y(t) = -\frac{g}{2}t^2 + \dot{y}_0 t + y_0. \tag{10.7}$$

Remark: You can also project $\mathbf{F} = m\mathbf{a}$ in the $\mathbf{E}_z$ direction, but since there are no forces in that direction and the initial velocity in that direction is 0, you conclude that the motion is planar.

10.2 Example: Projectile Motion with Viscous Drag

(Primer Section 1.5.3)

1. Same kinematics as above.
2. Force model — weight and Stokes drag:



$$\mathbf{W} = mg(-\mathbf{E}_y), \tag{10.8}$$

$$\mathbf{F}_D = -c_s\mathbf{v}. \tag{10.9}$$

The Stokes drag (1851) applies at low speeds in viscous fluids.

3. Balance of linear momentum:

$$-mg\mathbf{E}_y - c_s\left(\dot{x}\mathbf{E}_x + \dot{y}\mathbf{E}_y\right) = m\left(\ddot{x}\mathbf{E}_x + \ddot{y}\mathbf{E}_y\right). \tag{10.10}$$

Projecting:

$$-c_s\dot{x} = m\ddot{x}, \tag{10.11}$$

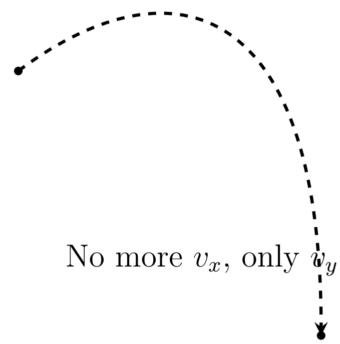
$$-mg - c_s\dot{y} = m\ddot{y}. \tag{10.12}$$

Given initial conditions $x_0, \dot{x}_0, y_0, \dot{y}_0$, these are an initial value problem (IVP).

4. Solving the x-equation (let $v_x = \dot{x}$):

$$v_x = v_{x_0} e^{-c_s t/m}, \quad \lim_{t\to\infty} v_x = 0. \tag{10.13}$$

Thus, the terminal velocity is only along $\mathbf{E}_y$.



Solving the y-equation:

$$v_y = -\frac{mg}{c_s} + \frac{1}{c_s}\left(mg + c_s v_{y_0}\right) e^{-c_s t/m}. \tag{10.14}$$

The terminal velocity: $\mathbf{v}_{\text{term}} = \frac{mg}{c_s}\left(-\mathbf{E}_y\right)$.

10.3 Example: Projectile Motion with Bluff Body Pressure Drag

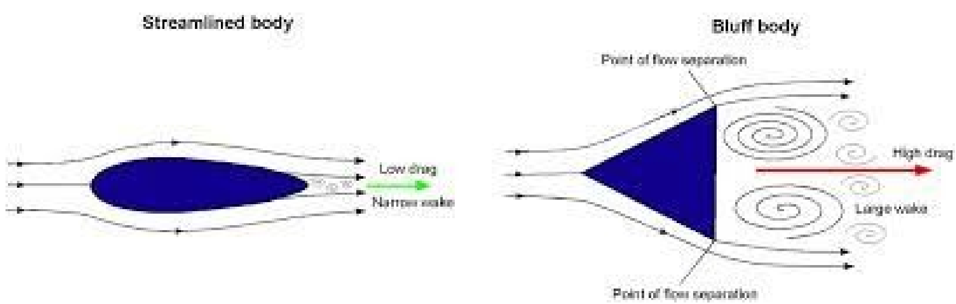


Figure 10.2: Bluff body pressure drag

$$\mathbf{F}_D = \frac{1}{2} m C_D v^2 \left(-\frac{\mathbf{v}}{\|\mathbf{v}\|}\right), \tag{10.15}$$

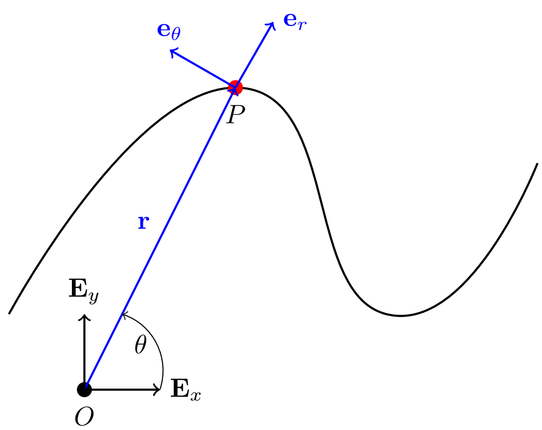
where C_D is the drag coefficient, ρ is the density of the fluid, and A is the projected area in the direction of motion.

See Section 1.6 in the book for a worked example.

11 Cylindrical Polar Coordinate System

The cylindrical polar coordinate system is three-dimensional. We start by introducing the two-dimensional polar basis, then extend it to three dimensions.

11.1 Polar Coordinates

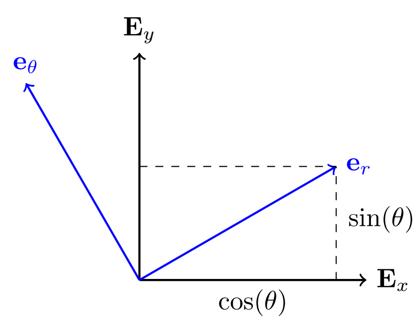


Consider $\mathbb{E}^2$. Define:

$$r = \sqrt{x^2 + y^2} \geq 0, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right). \tag{11.1}$$

So $x = r\cos(\theta)$ and $y = r\sin(\theta)$.

Define the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ as a rotation of $\{\mathbf{E}_x, \mathbf{E}_y\}$ by angle θ (positive CCW about $\mathbf{E}_z$):



> **Think!**
>
> **Question:** Express $\mathbf{e}_r$ and $\mathbf{e}_\theta$ in the Cartesian basis.
>
> **Answer**
>
> $$\mathbf{e}_r = \cos(\theta)\mathbf{E}_x + \sin(\theta)\mathbf{E}_y, \tag{11.2}$$
> $$\mathbf{e}_\theta = -\sin(\theta)\mathbf{E}_x + \cos(\theta)\mathbf{E}_y. \tag{11.3}$$

$\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is an orthonormal basis: $\mathbf{e}_r \cdot \mathbf{e}_r = 1$, $\mathbf{e}_\theta \cdot \mathbf{e}_\theta = 1$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$.

- $\mathbf{e}_r$ points in the direction of increasing r.
- $\mathbf{e}_\theta$ points in the direction of increasing θ.

> **Think!**
>
> **Question:** Calculate $\dot{\mathbf{e}}_r$ and $\dot{\mathbf{e}}_\theta$.
>
> **Answer**
>
> $$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r. \tag{11.4}$$

The position vector is $\mathbf{r} = r\mathbf{e}_r$ (with $r \geq 0$).

> **Think!**
>
> **Question:** Calculate $\mathbf{v}$ and $\mathbf{a}$.
>
> **Answer**
>
> $$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta,$$
> $$\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta.$$

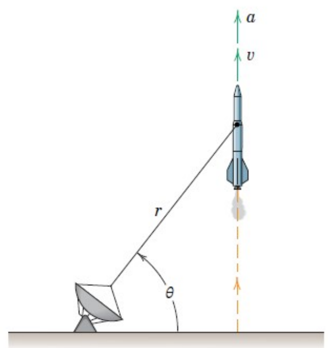
Note the components of acceleration:

- $r\ddot{\theta}$: tangential
- $\ddot{r}$: radial
- $r\dot{\theta}^2$: centripetal
- $2\dot{r}\dot{\theta}$: Coriolis

Remark: $a_r \neq \dot{v}_r$**.** Be careful: $v_r = \dot{r}$ so $\dot{v}_r = \ddot{r}$, but $a_r = \ddot{r} - r\dot{\theta}^2 \neq \dot{v}_r$ in general.

For rectilinear motions, Cartesian coordinates are usually easiest. For problems where r is directly measured (e.g. radar tracking), polar coordinates are natural.

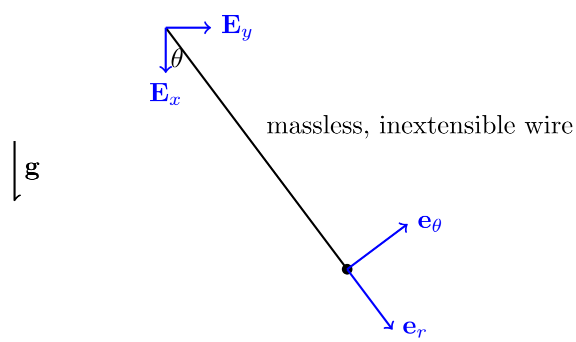
2/121 **SS** The rocket is fired vertically and tracked by the radar station shown. When θ reaches $60°$, other corresponding measurements give the values $r = 9$ km, $\ddot{r} = 21$ m/s^2, and $\dot{\theta} = 0.02$ rad/s. Calculate the magnitudes of the velocity and acceleration of the rocket at this position.



PROBLEM 2/121

11.1.1 Example: Nonlinear Pendulum

Consider a particle of mass m suspended from a rigid inextensible string of length ℓ. Find the equation of motion.



The constraint on the motion:

$$r = \ell = \text{const.} \implies \dot{r} = 0, \ \ddot{r} = 0. \tag{11.5}$$

Think!

Question: How many degrees of freedom does this system have?

Answer

One. A particle in the plane has 2 DOFs, but the constraint $r = \ell$ removes one. Knowing $\theta(t)$ completely describes the motion.

Following the four steps:

1. Choose polar coordinates:

$$\mathbf{r} = \ell \mathbf{e}_r, \quad \mathbf{v} = \ell\dot{\theta}\mathbf{e}_\theta, \quad \mathbf{a} = \ell\ddot{\theta}\mathbf{e}_\theta - \ell\dot{\theta}^2\mathbf{e}_r.$$

Orienting $\mathbf{e}_\theta$

$\mathbf{e}_\theta$ points in the direction of increasing θ, determined by the right-hand rule: point the thumb along $\mathbf{E}_z$; your fingers curl in the direction of increasing θ.

2. Forces:

$$\mathbf{T} = T\left(-\mathbf{e}_r\right), \quad \mathbf{W} = mg\left(\cos(\theta)\mathbf{e}_r - \sin(\theta)\mathbf{e}_\theta\right).$$

Remark: Tension

$\mathbf{T}$ is a constraint force keeping the particle on a circular path. If $T > 0$, the wire is in tension. The tension is the same throughout because the wire is massless.

3. Balance of linear momentum:

$$mg\mathbf{E}_x - T\mathbf{e}_r = m\ell\left(\ddot{\theta}\mathbf{e}_\theta - \dot{\theta}^2\mathbf{e}_r\right).$$

Projecting along $\mathbf{e}_\theta$ (hides unknown tension):

$$-mg\sin(\theta) = m\ell\ddot{\theta} \implies \ddot{\theta} + \frac{g}{\ell}\sin(\theta) = 0.$$

For small θ: $\ddot{\theta} + \frac{g}{\ell}\theta = 0$ (simple harmonic motion).

Projecting along $\mathbf{e}_r$ gives the tension: $T = m\left(\ell\dot{\theta}^2 + g\cos(\theta)\right)$.

11.2 Types of Accelerations Explained

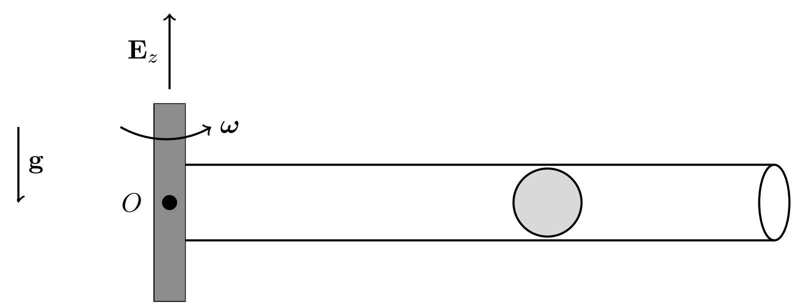
Recall:

$$\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta.$$

- **Radial** $\ddot{r}$: particle on a spring in rectilinear motion.
- **Centripetal** $-r\dot{\theta}^2$: particle on a circle at constant speed — the normal force steers the particle.
- **Coriolis** $2\dot{r}\dot{\theta}$: particle in a tube rotating at constant angular velocity.
- **Tangential** $r\ddot{\theta}$: particle pushed along a circle.

11.2.1 Example: Particle in a Rotating Tube

$\dot{\theta} = \omega = \text{const.}$, $\ddot{\theta} = 0$.



1. Kinematics:

$$\mathbf{r} = r\mathbf{e}_r, \quad \mathbf{v} = \dot{r}\mathbf{e}_r + r\omega\mathbf{e}_\theta, \quad \mathbf{a} = \left(\ddot{r} - r\omega^2\right)\mathbf{e}_r + 2\dot{r}\omega\mathbf{e}_\theta.$$

2. Assume no friction: $\mathbf{N} = N\mathbf{e}_\theta$.
3. Applying BoLM and projecting:

$$\left(\mathbf{F} = \dot{\mathbf{G}}\right)\cdot\mathbf{e}_r \implies \ddot{r} - \omega^2 r = 0 \quad \text{(EOM for } r(t)\text{)},$$
$$\left(\mathbf{F} = \dot{\mathbf{G}}\right)\cdot\mathbf{e}_\theta \implies N = 2m\dot{r}\omega \quad \text{(Coriolis reaction)}.$$

> **Coriolis acceleration**
>
> As the particle moves outward, $r\omega$ increases. The tube must supply the $\mathbf{e}_\theta$ acceleration $2\dot{r}\omega$ so the particle stays inside.

11.2.2 Example: Particle Pushed in a Circle

$\mathbf{P} = P\mathbf{e}_\theta$, $r = R = $ const.:

$$\begin{aligned}
\left(\mathbf{F} = \dot{G}\right) \cdot \mathbf{e}_\theta &\implies P = mR\ddot{\theta} \quad (\text{EOM for } \theta), \\
\left(\mathbf{F} = \dot{G}\right) \cdot \mathbf{e}_r &\implies N_r = mR\dot{\theta}^2 \quad (\text{centripetal}), \\
\left(\mathbf{F} = \dot{G}\right) \cdot \mathbf{E}_z &\implies N_z = mg \quad (\text{vertical equilibrium}).
\end{aligned}$$

11.3 Cylindrical-Polar Coordinates

In $\mathbb{E}^3$, the polar system extends by adding z:

$$\begin{bmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{E}_z \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{E}_x \\ \mathbf{E}_y \\ \mathbf{E}_z \end{bmatrix}.$$

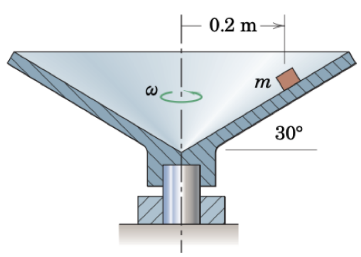
Position, velocity, acceleration:

$$\begin{aligned}
\mathbf{r} &= r\mathbf{e}_r + z\mathbf{E}_z, \\
\mathbf{v} &= \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{E}_z, \\
\mathbf{a} &= \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\mathbf{e}_\theta + \ddot{z}\mathbf{E}_z.
\end{aligned}$$

11.3.1 Example: Particle on a Cone

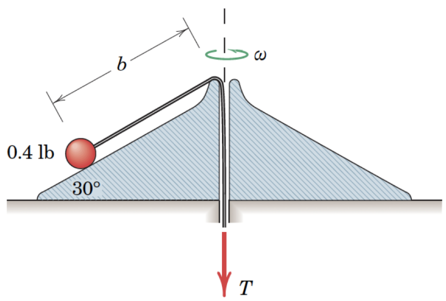
See animation for 03/067 here.

3/67 The small object is placed on the inner surface of the conical dish at the radius shown. If the coefficient of static friction between the object and the conical surface is 0.30, for what range of angular velocities ω about the vertical axis will the block remain on the dish without slipping? Assume that speed changes are made slowly so that any angular acceleration may be neglected.



PROBLEM 3/67

3/193 **SS** The 0.4-lb ball and its supporting cord are revolving about the vertical axis on the fixed smooth conical surface with an angular velocity of 4 rad/sec. The ball is held in the position $b = 14$ in. by the tension T in the cord. If the distance b is reduced to the constant value of 9 in. by increasing the tension T in the cord, compute the new angular velocity ω and the work $U'_{1\text{-}2}$ done on the system by T.



PROBLEM 3/193

12 Constrained Motion

The **degree of freedom (DOF)** of a system is the number of independent coordinates needed to describe a configuration of the system.

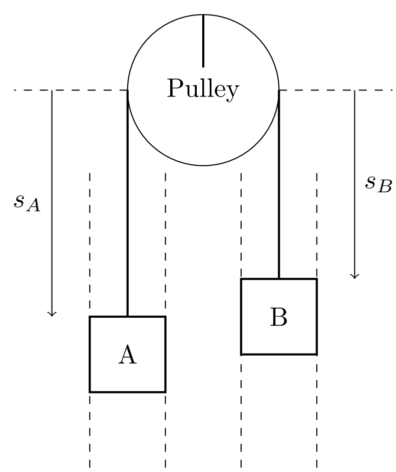
Examples:

- A particle free to move in $\mathbb{E}^2$ has 2 DOFs.
- A particle constrained to move on a curve in $\mathbb{E}^2$ has 1 DOF.
- Two particles free to move in $\mathbb{E}^2$ have a combined total of 4 DOFs.
- If two particles in $\mathbb{E}^2$ are constrained to move on the same curve, they have 2 DOFs.
- If two particles in $\mathbb{E}^2$ are connected by a rigid rod, they have 3 DOFs. The constraint $\left\|\mathbf{r}_{B/A}\right\| = l \,\forall t$ provides a relationship between their coordinates that eliminates one DOF.
- If two particles in $\mathbb{E}^2$ are connected by a rigid rod and constrained to move on a curve, they have 1 DOF.

Example: Consider two particles in smooth vertical slots. If A and B move independently, the system has 2 DOFs. If connected by an inextensible cable, the system has 1 DOF.

The inextensibility constraint $\ell = \text{const.}$ can be written as:

$$\ell = s_A + c + s_B \implies \Delta s_B = -\Delta s_A \implies \dot{s}_B = -\dot{s}_A. \tag{12.1}$$



See Set 06: Constrained Motion Problem 02/172.

13 The Serret-Frenet Basis

In general, the path of a particle is a curve in space — we refer to such curves as *space curves*.

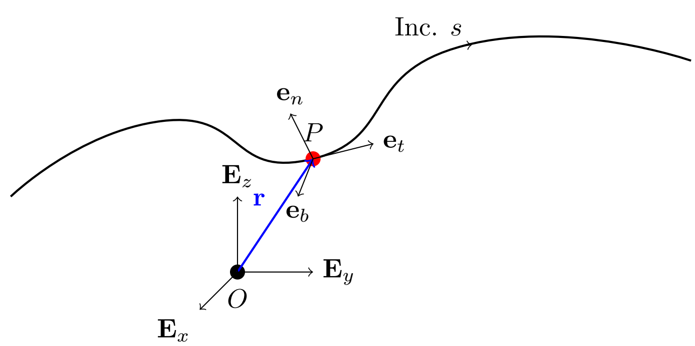
Consider a fixed curve $\mathcal{C}$ embedded in $\mathbb{E}^3$. Let the position vector of a point $P \in \mathcal{C}$ be $\mathbf{r}$. In Cartesian coordinates, $\mathbf{r} = x\mathbf{E}_x + y\mathbf{E}_y + z\mathbf{E}_z$.

Recall the arc-length parameter s associated with $\mathcal{C}$:

$$\left(\frac{ds}{dt}\right)^2 = \frac{d\mathbf{r}}{dt} \cdot \frac{d\mathbf{r}}{dt}.$$

This parameter uniquely identifies a point P on $\mathcal{C}$, giving the representation $\mathbf{r} = \hat{\mathbf{r}}(s)$.

13.1 The Serret-Frenet Triad



The Serret-Frenet basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ is defined at the point P; it depends on the curve and the particle's location on it.

Think!

Question: How do we define the unit tangent vector?

Answer

...

Recall the unit tangent vector:

$$\mathbf{e}_t = \hat{\mathbf{e}}_t(s) = \frac{d\mathbf{r}}{ds} = \lim_{\Delta s \to 0} \frac{\hat{\mathbf{r}}(s + \Delta s) - \hat{\mathbf{r}}(s)}{\Delta s}. \tag{13.1}$$

Think!

Question: What do we know about $\frac{d\hat{\mathbf{e}}_t}{ds}$?

Answer

We can show that $\mathbf{e}_t$ and $\frac{d\hat{\mathbf{e}}_t}{ds}$ are perpendicular by differentiating $\mathbf{e}_t \cdot \mathbf{e}_t = 1$:

$$\mathbf{e}_t \cdot \frac{d\mathbf{e}_t}{ds} = 0. \tag{13.2}$$

Note! Derivatives of a unit vector

It is a common mistake to assume that the derivative of a unit vector is necessarily a unit vector. Recall: $\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$ is not generally a unit vector.

Hence, we define:

$$\kappa \mathbf{e}_n = \frac{d\mathbf{e}_t}{ds}, \tag{13.3}$$

where $\kappa = \left\| \frac{d\mathbf{e}_t}{ds} \right\| \geq 0$ and $\|\mathbf{e}_n\| = 1$.

- $\mathbf{e}_n$ is the unit **principal normal** vector.
- κ is the **curvature** of $\mathcal{C}$ at P.
- The **radius of curvature** is $\rho = 1/\kappa$.

For any point on a curve, three nearby non-collinear points define the **osculating circle** (radius ρ, center of curvature).

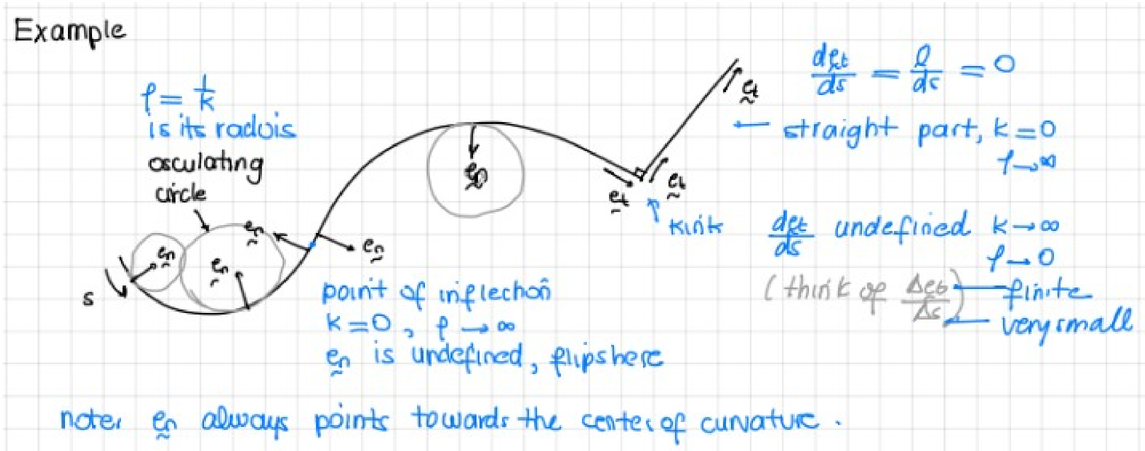


Figure 13.1: Serret-Frenet triad example

Think!

Question: When is $\mathbf{e}_n$ not uniquely defined?

Answer

When $d\mathbf{e}_t/ds = \mathbf{0}$ (i.e. $\kappa = 0$), such as on a straight line, at an inflection point, or at a corner. In this case, $\mathbf{e}_n$ is defined as any unit vector perpendicular to $\mathbf{e}_t$.

The unit **binormal** vector:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n.$$

Concluding remarks:

- $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ is orthonormal and right-handed.
- The **osculating plane** is spanned by $\mathbf{e}_t$ and $\mathbf{e}_n$; it contains the osculating circle.
- The **rectifying plane** is spanned by $\mathbf{e}_t$ and $\mathbf{e}_b$.
- The **normal plane** is spanned by $\mathbf{e}_n$ and $\mathbf{e}_b$.

Check out this animation showing the osculating circle, and this video placing the Serret-Frenet basis on a bobsled.

13.2 Kinematics

Position, velocity, and acceleration in the Serret-Frenet basis:

$$\mathbf{v} = \frac{ds}{dt}\mathbf{e}_t = v\mathbf{e}_t,$$
$$\mathbf{a} = \dot{v}\mathbf{e}_t + \kappa v^2 \mathbf{e}_n.$$

13.2.1 Example: Problem 02/279

Draw the Serret-Frenet basis. Indicate whether the car is accelerating or decelerating and its direction of motion.

13.3 Kinetics

$$\mathbf{F} = m\mathbf{a} \tag{13.4}$$
$$F_t\mathbf{e}_t + F_n\mathbf{e}_n + F_b\mathbf{e}_b = m\left(\dot{v}\mathbf{e}_t + \kappa v^2\mathbf{e}_n\right). \tag{13.5}$$

Note that forces in the binormal direction should always balance.

13.3.1 Example: Problem 03/040 and Roller Coaster Design

Think!

Question: Consider the design of a roller coaster that is a circular ring. Comment on its safety.

Answer

At the transition between the circular and straight portions, there is a jump in curvature, hence a jump in acceleration, hence a jump in the normal force — producing a jerk. For a circular loop, the normal force goes from 0 to a finite value instantaneously. This could snap the rider's neck. Instead, we use two connected **clothoids** (Euler spirals), whose curvature is proportional to arc length.

13.4 The Serret-Frenet Formulae

We want the derivatives of the Serret-Frenet basis vectors with respect to s.

We already have:

$$\frac{d\mathbf{e}_t}{ds} = \kappa \mathbf{e}_n. \tag{13.6}$$

Since $\mathbf{e}_b \cdot \mathbf{e}_b = 1$, we have $\frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_b = 0$. Also from $\mathbf{e}_t \cdot \mathbf{e}_b = 0$, we get $\frac{d\mathbf{e}_b}{ds} \cdot \mathbf{e}_t = 0$. So $\frac{d\mathbf{e}_b}{ds}$ is parallel to $\mathbf{e}_n$:

$$\frac{d\mathbf{e}_b}{ds} = -\tau \mathbf{e}_n, \tag{13.7}$$

where τ is the **torsion** of $\mathcal{C}$ at P.

Then:

$$\frac{d\mathbf{e}_n}{ds} = -\kappa \mathbf{e}_t + \tau \mathbf{e}_b. \tag{13.8}$$

In summary (the Serret-Frenet formulae):

$$\begin{bmatrix} \frac{d\mathbf{e}_t}{ds} \\ \frac{d\mathbf{e}_n}{ds} \\ \frac{d\mathbf{e}_b}{ds} \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} \mathbf{e}_t \\ \mathbf{e}_n \\ \mathbf{e}_b \end{bmatrix}. \tag{13.9}$$

Define the **Darboux vector** $\omega_{SF} = \kappa \mathbf{e}_b + \tau \mathbf{e}_t$ so that:

$$\frac{d\mathbf{e}_i}{ds} = \omega_{SF} \times \mathbf{e}_i, \quad i = t, n, b. \tag{13.10}$$

13.4.1 Example: A Particle on a Helix

For a helix $x = R\cos(\theta)$, $y = R\sin(\theta)$, $z = \alpha R\theta$:

$$\kappa = \frac{1}{R(1+\alpha^2)}, \qquad \tau = \frac{\alpha}{(1+\alpha^2)R}. \tag{13.11}$$

13.5 The Curvature Formula for a Plane Curve

For a plane curve $y = f(x)$, $z = z_0$:

$$\kappa = \kappa(x) = \frac{\left|\frac{d^2 f}{dx^2}\right|}{\left(\sqrt{1 + \left(\frac{df}{dx}\right)^2}\right)^3}. \tag{13.12}$$

13.6 Alternative Notation for Plane Curves

This section is not usually covered.

For plane curves, some texts define the angle $\beta = \beta(s)$ such that:

$$\mathbf{e}_t = \cos(\beta)\mathbf{E}_x + \sin(\beta)\mathbf{E}_y, \qquad \bar{\mathbf{e}}_n = \cos(\beta)\mathbf{E}_y - \sin(\beta)\mathbf{E}_x. \tag{13.13}$$

Then $\kappa = \left|\frac{d\beta}{ds}\right|$ and $\frac{d\beta}{ds}$ is the rate of rotation of the triad about $\mathbf{E}_z$.

14 Spring Force

14.1 Introduction

Think!

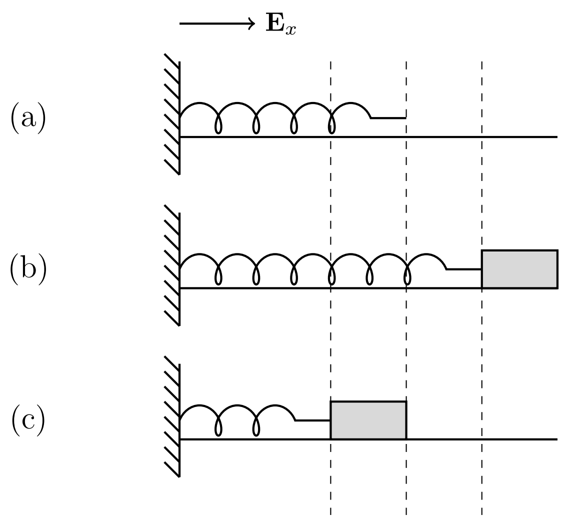
Question: What does it mean for a spring to be linear?

Answer

Consider a mass m suspended by a spring causing an extension δ. Suspending a mass $2m$ causes double the extension, 2δ.

Hooke's Law: The force generated by a spring is assumed to be linearly proportional to its extension/compression.

Consider a block of mass m attached to a linear spring of stiffness k and unstretched length ℓ_0. Denote the current length by ℓ and the stretch by $\varepsilon = \ell - \ell_0$.



Think!

Question: What is the stretch in each case? What is the spring force?

Answer

(a) Unstretched: $\varepsilon = 0$, no spring force.

(b) Extended: $\varepsilon = \ell - \ell_0 > 0$. Spring force $\mathbf{F}_s = -k\varepsilon\mathbf{E}_x$ (pointing left, toward equilibrium).

(c) Compressed: $\varepsilon = \ell - \ell_0 < 0$. Spring force $\mathbf{F}_s = -k\varepsilon\mathbf{E}_x$ (pointing right, toward equilibrium).

! Note! Prescribing the spring force

The spring force always seeks to return the particle to the unstretched state. The same expression applies whether the spring is extended or compressed.

14.2 Formalism

In general:

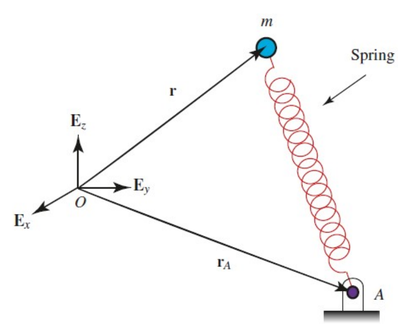


Figure 14.1: General spring configuration

Think!

Question: What is the prescription of the spring force in this general case?

Answer

...

The stretch is:

$$\varepsilon = \|\mathbf{r} - \mathbf{r}_A\| - \ell_0, \tag{14.1}$$

where ℓ_0 is the unstretched length. The spring force is:

$$\mathbf{F}_s = -K\left(\|\mathbf{r} - \mathbf{r}_A\| - \ell_0\right)\frac{\mathbf{r} - \mathbf{r}_A}{\|\mathbf{r} - \mathbf{r}_A\|}. \tag{14.2}$$

This direction is correct in both tension and compression.

14.2.1 Example: Horizontal Simple Harmonic Oscillator

Derive the equations of motion with the origin at the fixed end of the spring.

Think!

Question: How does the equation of motion change if the origin is taken at the free end when the spring is unstretched?

Answer

The equation of motion is

$$\ddot{x} + \frac{k}{m}x = 0, \tag{14.3}$$

where x is measured from the undeformed position.

14.2.2 Example: Vertical Simple Harmonic Oscillator

By analogy, the equation of motion of a vertical harmonic oscillator is:

$$\ddot{x} + \frac{k}{m}x = 0, \tag{14.4}$$

where x is measured from the **static equilibrium** position.

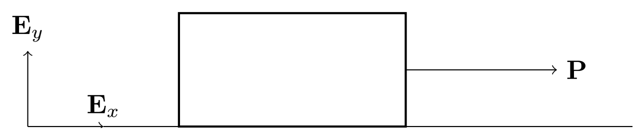
See this video for intuition on static deflection.

15 Friction Force

Most theories of friction arise from studies by the French scientist Charles Augustin de Coulomb (1736–1806).

15.1 Introduction

Consider a block on a rough horizontal surface pushed with force $\mathbf{P} = P\mathbf{E}_x$.



Upon increasing P, we observe:

- For small P, the block remains at rest.
- Beyond a critical value $P = P^*$, the block starts to move.
- Once in motion, a constant value $P = P^{**}$ is required to maintain constant speed.
- Both P^* and P^{**} are proportional to the normal force $\|\mathbf{N}\|$.

15.2 Formalism: Coulomb's Theory of Friction

We distinguish two types of friction: static and kinetic.

15.2.1 Static Friction

When the relative velocity is zero ($\mathbf{v}_{rel} = \mathbf{0}$), the friction force $\mathbf{F}_f$ is **unknown** and lies along the contact surface. We must solve for both its magnitude and direction. We say the static friction force points in the direction that **impedes motion**.

Think!

Question: Consider a block on a rough inclined surface that is not moving. Draw the FBD and write the equation of motion. Do you have as many equations as unknowns?

Answer

Yes. The friction and normal forces have unknown magnitudes, but the block's acceleration is known (it matches the surface acceleration). There are as many equations as unknowns.

The static friction criterion:

$$\left\|\mathbf{F}_f\right\| \leq \mu_s \left\|\mathbf{N}\right\|, \tag{15.1}$$

where μ_s is the static friction coefficient.

15.2.2 Kinetic Friction

When two bodies are in relative motion ($\mathbf{v}_{rel} \neq \mathbf{0}$), the friction force is **known**:

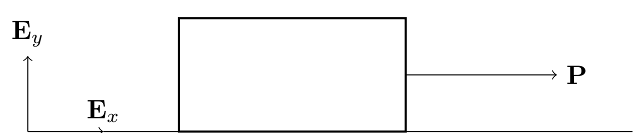
$$\mathbf{F}_f = \mu_k \left\|\mathbf{N}\right\| \left(-\frac{\mathbf{v}_{rel}}{\left\|\mathbf{v}_{rel}\right\|}\right), \tag{15.2}$$

where μ_k is the kinetic friction coefficient.

Remarks:

- μ_k is also written μ_d (dynamic friction coefficient).
- Kinetic friction is a prescription, not a constraint force.
- The magnitude of kinetic friction does not depend on speed.

15.2.3 Example: Crate Pushed with Incremental Force



1. Kinematics: $\mathbf{r} = x\mathbf{E}_x + y_0\mathbf{E}_y$, $\mathbf{v} = \dot{x}\mathbf{E}_x$, $\mathbf{a} = \ddot{x}\mathbf{E}_x$.
2. FBD with forces $\mathbf{P} = P\mathbf{E}_x$, $\mathbf{W} = -mg\mathbf{E}_y$, $\mathbf{N} = N\mathbf{E}_y$, $\mathbf{F}_f = F_f(-\mathbf{E}_x)$.

   When kinetic: $\mathbf{F}_f = -\mu_k \left\|\mathbf{N}\right\| \operatorname{sgn}(\dot{x})\mathbf{E}_x$.

3. BoLM:

$$P\mathbf{E}_x - mg\mathbf{E}_y + N\mathbf{E}_y - F_f\mathbf{E}_x = m\ddot{x}\mathbf{E}_x. \tag{15.3}$$

4. Initially at rest (static friction). Setting $\ddot{x} = 0$:

$$P = F_f, \quad N = mg. \tag{15.4}$$

   The block stays at rest while $P \leq \mu_s mg$. The maximum static value is $P^* = \mu_s mg$.

Once sliding, $\mathbf{F}_f = -\mu_k mg\mathbf{E}_x$ and:

$$\ddot{x} = \frac{P}{m} - \mu_k g. \tag{15.5}$$

For constant speed, $P^{**} = \mu_k mg$.

15.2.4 Example: A Box on a Ramp

(MKB problem 03/002.)

When it is unknown whether the block is in motion:

1. Assume sticking: $\mathbf{v}_{rel} = \mathbf{0}$.
2. Calculate the static friction force.
3. Check the static friction criterion.
    - If satisfied: the block sticks. Analysis complete.
    - If not satisfied: the block slides. Apply kinetic friction and resolve.

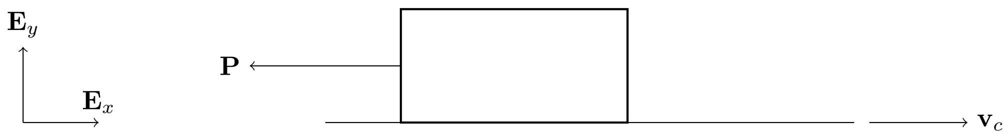
15.3 Relative Velocity

A common misconception: if an object is moving, the friction acting on it must be kinetic. Counter-example: a crate on a moving truck. If the crate moves with the truck (same velocity), the friction between them is **static**. Only if the crate slides on the truck bed is the friction kinetic.

Similar examples include a person in an elevator and a person riding a bus.

15.3.1 Example: Nonzero Relative Velocity

- **Box held fixed on a moving conveyor belt:** The belt moves, so $\mathbf{v}_{rel} = -\mathbf{v}_c \neq \mathbf{0}$ — friction is kinetic. The box is stationary in space but moving relative to the belt.



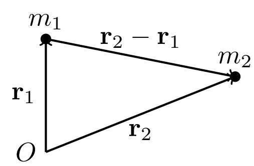
- **Person standing in an accelerating bus:** Friction is static. The impending motion of the person (relative to the bus) is toward the rear, so the static friction force points toward the front.
- **Car rolling without slipping:** Friction is static and points in the direction of motion, opposing the tendency of the tire to slip backward.

15.4 Normal and Friction Force Prescription

See this video.

- For a particle moving on a **space curve**: $\mathbf{N} = N_n\mathbf{e}_n + N_b\mathbf{e}_b$ and $\mathbf{F}_f = F_f\mathbf{e}_t$.
- For a particle moving on a **surface** with normal $\mathbf{n}$ and tangent directions $\mathbf{t}_1$, $\mathbf{t}_2$: $\mathbf{N} = N\mathbf{n}$ and $\mathbf{F}_f = F_{f1}\mathbf{t}_1 + F_{f2}\mathbf{t}_2$.

16 Gravitational Force Model

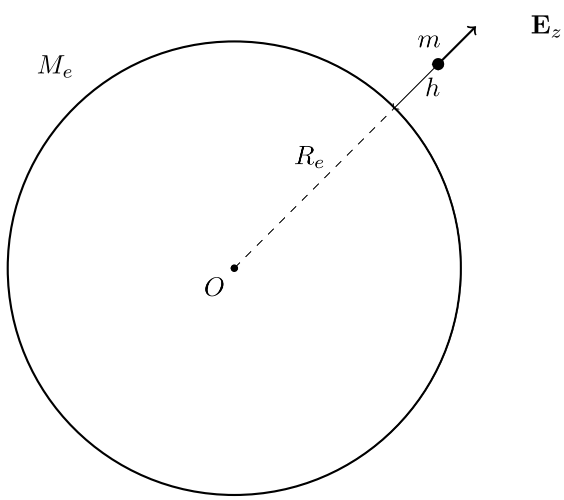


Call $\mathbf{r}_2 - \mathbf{r}_1 = \mathbf{r}_{2/1}$. According to Newton's law of gravitation, the force on m_1 due to m_2 is:

$$\mathbf{F}_1 = G\frac{m_1 m_2}{\left\|\mathbf{r}_{2/1}\right\|^3}\mathbf{r}_{2/1}, \tag{16.1}$$

where $G = 6.67408 \times 10^{-11}$ m^3 kg^{-1} s^{-2} is the gravitational constant.

16.1 Example: Attraction Between Earth and an Object of Mass m



$$\mathbf{F} = G\frac{M_e m}{(R_e + h)^2}\left(-\mathbf{E}_z\right). \tag{16.2}$$

Since $h \ll R_e$, we have $(R_e + h)^2 \approx R_e^2$, giving:

$$\mathbf{F} = G\frac{M_e}{R_e^2}m\left(-\mathbf{E}_z\right) = mg\left(-\mathbf{E}_z\right), \tag{16.3}$$

where $g = GM_e/R_e^2 = 9.81$ m/s^2 $= 32.2$ ft/s^2.

Earth's parameters:

- $M_e = 5.972 \times 10^{24}$ kg
- $R_e = 6.371 \times 10^{6}$ m

17 Power, Work, and Energy

17.1 Power and Work

The **mechanical power** of a force $\mathbf{F}$ acting on a particle with absolute velocity $\mathbf{v}$ is:

$$\mathcal{P} = \mathbf{F} \cdot \mathbf{v}. \tag{17.1}$$

The **work** done by $\mathbf{F}$ over the interval $[t_1, t_2]$ is:

$$W_{\mathbf{F},12} = \int_{t_1}^{t_2} \mathbf{F} \cdot \mathbf{v}\, dt = \int_{t_1}^{t_2} \mathbf{F} \cdot d\mathbf{r}, \tag{17.2}$$

since $d\mathbf{r} = \mathbf{v}\, dt$. So power is the derivative of work: $\mathcal{P} = \dot{W}$.

In SI units: work is in Newton meters (Joules) and power in Watts.

Notes:

- There are many representations of this integral depending on the coordinate system:
  - Cartesian: $d\mathbf{r} = dx\mathbf{E}_x + dy\mathbf{E}_y + dz\mathbf{E}_z$
  - Polar: $d\mathbf{r} = dr\mathbf{e}_r + r\, d\theta\mathbf{e}_\theta$
  - Serret-Frenet: $d\mathbf{r} = ds\, \mathbf{e}_t$
- This integral is **path dependent** — we need to know the particle's path to compute it.

Think!

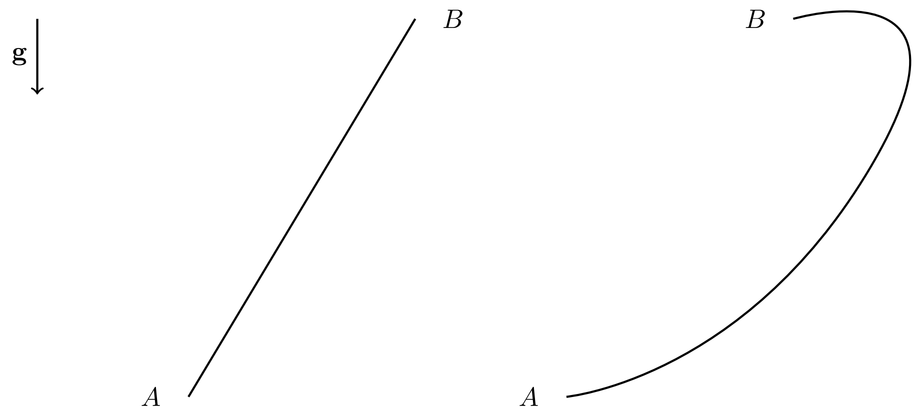
Question: In what case would the work of a force be zero?

Answer

The work is zero if the force is perpendicular to the displacement/velocity, or if the force is applied at a fixed point.

17.1.1 Example: Particle on a Rail

Consider two points A and B in the vertical plane connected by a smooth rail.



Think!

Question: What is the work done by forces on an object traveling from A to B on (a) a straight rail or (b) a curved path?

Answer

In both cases $\mathbf{F} = \mathbf{W} + \mathbf{N}$. Since $\mathbf{N} \perp d\mathbf{r}$, the normal force does no work. The work of weight is:

$$W_{\mathbf{W},1-2} = \int mg\mathbf{E}_y \cdot d\mathbf{r} = mg\mathbf{E}_y \cdot \Delta\mathbf{r}_{1-2}.$$

$\Delta\mathbf{r}_{1-2}$ is the same for both rails, so weight does the same work regardless of path.

Think!

Question: What changes if the rail is rough?

Answer

With friction:

$$W_{1-2} = W_{\mathbf{W},1-2} - \mu_k \int_{s_1}^{s_2} \|\mathbf{N}\| \, ds.$$

Friction work is path dependent and dissipative ($W_{\mathbf{F}_f,1-2} < 0$).

17.2 Kinetic Energy

The **kinetic energy** of a particle:

$$T = \frac{m}{2}\mathbf{v} \cdot \mathbf{v} = \frac{1}{2}mv^2. \tag{17.3}$$

In different bases:

$$T = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2).$$

17.3 Energy

Energy is defined as the ability to perform work.

17.4 The Work-Energy Theorem

The rate of change of kinetic energy equals the mechanical power of the resultant force:

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v} = \mathcal{P}. \tag{17.4}$$

Proof:

$$\frac{d}{dt}T = \frac{d}{dt}\left(\frac{1}{2}m\mathbf{v} \cdot \mathbf{v}\right) = m\mathbf{a} \cdot \mathbf{v} = \mathbf{F} \cdot \mathbf{v}.$$

Integral form:

$$T_2 - T_1 = W_{\mathbf{F},12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r}. \tag{17.5}$$

17.5 Conservative Forces

Forces whose work depends only on the endpoints of a path are **conservative**. Weight (constant force) is conservative; friction is not.

A force $\mathbf{F}_c$ is conservative if there exists a scalar **potential energy** $U = U(\mathbf{r})$ such that:

$$\mathbf{F}_c = -\frac{\partial U}{\partial \mathbf{r}} = -\nabla U. \tag{17.6}$$

Then:

$$W_{1-2} = -U_2 + U_1 = -\Delta U. \tag{17.7}$$

If potential energy decreases ($U_2 \leq U_1$), the work is positive and kinetic energy increases.

17.5.1 Constant Forces

Constant force $\mathbf{C}$ is conservative with potential $U = -\mathbf{C} \cdot \mathbf{r}$.

The weight near Earth's surface: $U_{\mathbf{W}} = mgy$.

17.5.2 Spring Force

The spring force is conservative with potential:

$$U_s = \frac{1}{2} K \varepsilon^2.$$

$U_s > 0$ in both compression and tension — it represents the spring's capacity to do work.

17.5.3 Gravitational Force

The gravitational force is conservative with:

$$U = -\frac{G M_e m}{r}.$$

Drag, friction, tension in inextensible cables, and normal forces are nonconservative.

17.6 Energy and Its Conservation

If only conservative forces act, define total mechanical energy $E = T + U$. Then:

$$\dot{T} = -\dot{U} \implies \dot{E} = \dot{T} + \dot{U} = 0.$$

E is conserved when all work-doing forces are conservative.

If nonconservative forces also act:

$$E_B - E_A = W_{\mathbf{F}_{nc}, AB}. \tag{17.8}$$

17.6.1 Example: Smooth vs. Rough Rail

Smooth rail: $\mathbf{F} \cdot \mathbf{v} = \mathbf{W} \cdot \mathbf{v} + \mathbf{N} \cdot \mathbf{v} = \mathbf{W} \cdot \mathbf{v} = -\dot{U}$, so $\dot{E} = 0$.

Rough rail: $\dot{E} = \mathbf{F}_f \cdot \mathbf{v} < 0$, so energy bleeds off over time. Integrating:

$$E(t_2) = E(t_1) + W_{\mathbf{F}_f,1-2} < E(t_1).$$

18 Momenta and Impulses of Particles

18.1 Linear Momentum and Its Conservation

Recall the linear momentum $\mathbf{G} = m\mathbf{v}$.

18.1.1 Linear Impulse and Linear Momentum

The integral form of the balance of linear momentum:

$$\mathbf{G}(t_1) - \mathbf{G}(t_0) = \int_{t_0}^{t_1} \mathbf{F}\, dt. \tag{18.1}$$

The time integral of a force is its **linear impulse**. This form is more general than $\mathbf{F} = m\mathbf{a}$ because it does not require $\mathbf{v}$ to be differentiable.

18.1.2 Conservation of Linear Momentum

Think!

Question: Under what conditions is the linear momentum of a system conserved in a direction $\mathbf{c}$?

Answer

$$\frac{d}{dt}(\mathbf{G} \cdot \mathbf{c}) = \mathbf{F} \cdot \mathbf{c} + \mathbf{G} \cdot \dot{\mathbf{c}} = 0. \tag{18.2}$$

For a constant $\mathbf{c}$, $\mathbf{G} \cdot \mathbf{c}$ is conserved if and only if $\mathbf{F} \cdot \mathbf{c} = 0$ — i.e., no force in that direction.

18.1.3 Example

Example

Question: Is the linear momentum of a projectile conserved? In a certain direction?

Answer

With $\mathbf{W} = -mg\mathbf{E}_y$, linear momentum is conserved in the $\mathbf{E}_x$ direction (no force there) but not in $\mathbf{E}_y$.

18.2 The Moment of a Force

Think!

Question: How do you calculate the moment of a force about a point?

Answer

The moment of force $\mathbf{F}$ applied at A about point P is:

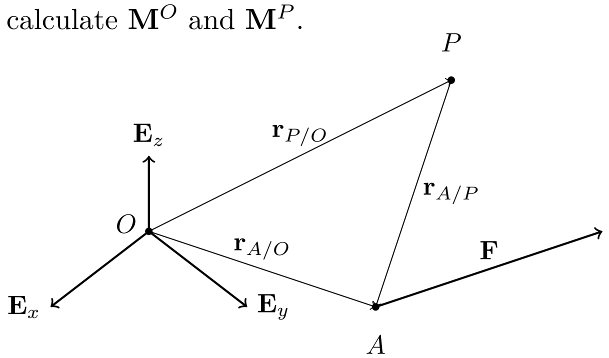
$$\mathbf{M}^P = (\mathbf{r}_A - \mathbf{r}_P) \times \mathbf{F}. \tag{18.3}$$

Example

Question: Given

$$\begin{aligned}\mathbf{F} &= F_x\mathbf{E}_x + F_y\mathbf{E}_y + F_z\mathbf{E}_z,\\ \mathbf{r}_P &= -2\mathbf{E}_x + 3\mathbf{E}_y + \mathbf{E}_z,\\ \mathbf{r}_A &= -\mathbf{E}_x + 2\mathbf{E}_y - \mathbf{E}_z,\end{aligned}$$

calculate $\mathbf{M}^O$ and $\mathbf{M}^P$.



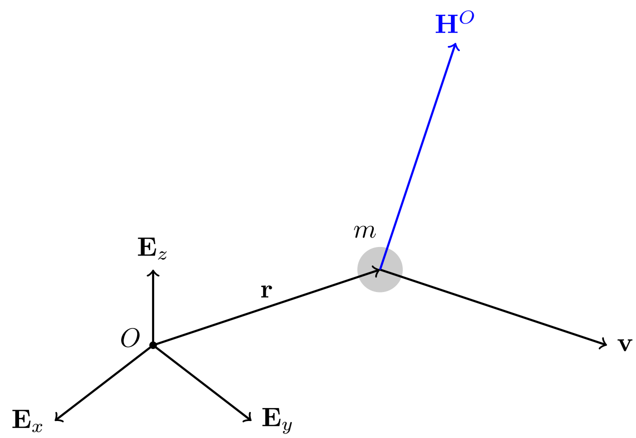
Answer

...

18.3 Angular Momentum and Its Conservation

Let $\mathbf{r}$ be the position vector of a particle relative to a fixed point O, and $\mathbf{v}$ its absolute velocity. The **angular momentum** relative to O:

$$\mathbf{H}^O = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{G}. \tag{18.4}$$



18.3.1 Angular Momentum Theorem

From the BoLM:

$$\dot{\mathbf{H}}^O = \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) = \underbrace{\mathbf{v} \times m\mathbf{v}}_{\mathbf{0}} + \mathbf{r} \times m\dot{\mathbf{v}} = \mathbf{r} \times \mathbf{F}. \tag{18.5}$$

So $\dot{\mathbf{H}}^O = \mathbf{M}^O$ (the angular momentum theorem).

18.3.2 Conservation of Angular Momentum

Think!

Question: Under what conditions is $\mathbf{H}^O \cdot \mathbf{c}$ conserved?

Answer

For a constant $\mathbf{c}$: if and only if $(\mathbf{r} \times \mathbf{F}) \cdot \mathbf{c} = 0$.

18.3.3 Central Force Problems

A **central force** problem has $\mathbf{F}$ parallel to $\mathbf{r}$, so $\dot{\mathbf{H}}^O = \mathbf{0}$ and $\mathbf{H}^O = h\mathbf{h} = \text{constant}$.

The vectors $\mathbf{r}$ and $\mathbf{v}$ remain in a fixed plane through O. Choosing $\mathbf{E}_z = \mathbf{h}$:

$$\mathbf{H}^O = h\mathbf{E}_z = \mathbf{r}(t_0) \times m\mathbf{v}(t_0). \tag{18.6}$$

18.3.4 Kepler's Problem

The gravitational force on a planet of mass m by the sun of mass M:

$$\mathbf{F} = -\frac{GmM}{\|\mathbf{r}\|^3}\mathbf{r}, \qquad U = -\frac{GmM}{\|\mathbf{r}\|}. \tag{18.7}$$

Two conserved quantities:

$$E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{GMm}{r}, \tag{18.8}$$

$$h = \mathbf{H}^O \cdot \mathbf{E}_z = mr^2\dot{\theta}. \tag{18.9}$$

Watch this video on Kepler's laws.

18.3.5 Particle on a Smooth Cone

Show that $\mathbf{H}^O \cdot \mathbf{E}_z$ is conserved.

19 Collisions

19.1 Collision of Two Bodies

Think!

Question: What happens in reality when two cars collide?

Answer

In reality, colliding bodies deform — possibly permanently — and their rotational inertia may change.

Ignoring rotational motion, the simplest model uses a mass particle for each body. Since particles do not deform, we introduce the **coefficient of restitution** e.

19.2 Frictionless, Oblique, Central Impact of Two Bodies

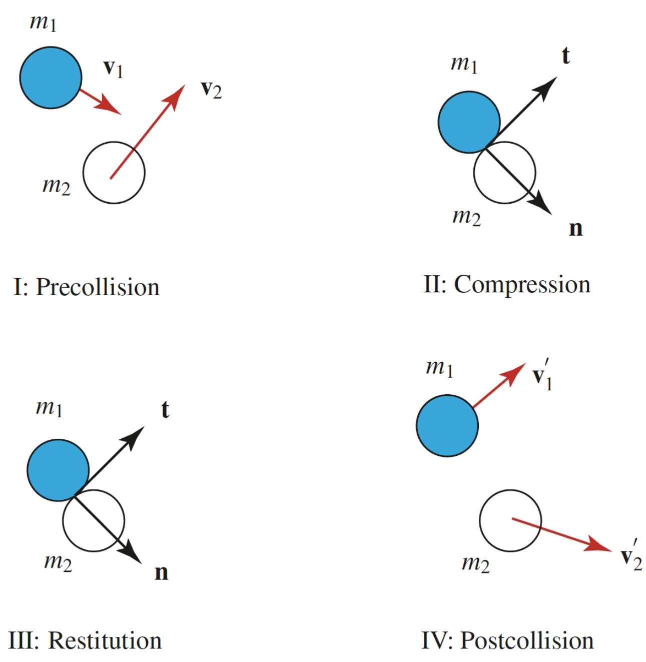


Fig. 6.4 *Four stages of a collision.*

Figure 19.1: Figure from O. M. O'Reilly's Dynamics Primer

19.2.1 Assumptions

- Model each body as a mass particle.
- We ignore the rotational inertias of bodies (purely translational motion).
- All points on a body have the same velocity.
- No friction between particles during collision — all impact forces are along $\mathbf{n}$.
- The duration of compression $(t_1 - t_0)$ and restitution $(t_2 - t_1)$ are very small.
- At time t_1, both particles have the same velocity along $\mathbf{n}$: v_{II}.
- $\mathbf{F}_{1C} = -\mathbf{F}_{2C}$, $\mathbf{F}_{1R} = -\mathbf{F}_{2R}$.

19.2.2 Impact Stages

Let $\mathbf{n}$ be the common unit normal between the bodies at contact. Define $\{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ as an orthonormal set. These vectors are assumed constant for the duration of impact.

19.2.3 Linear Impulses During Impact

Think!

Question: Draw the FBDs of m_1 and m_2 separately during compression and restitution.

Answer

- $\mathbf{F}_{1_d} = F_{1_d}\mathbf{n}$: force exerted by body 2 on body 1 during compression.
- $\mathbf{F}_{1_r} = F_{1_r}\mathbf{n}$: force exerted by body 2 on body 1 during restitution.
- $\mathbf{F}_{2_d}$, $\mathbf{F}_{2_r}$: reaction forces by body 1 on body 2.
- All other forces have resultants $\mathbf{R}_1$ and $\mathbf{R}_2$.

Since the collision duration is very small, the impulses of $\mathbf{R}_1$ and $\mathbf{R}_2$ vanish. Approximating:

$$\int_{t_0}^{t_1} \left(\mathbf{F}_{1_d} + \mathbf{R}_1\right) d\tau \approx \int_{t_0}^{t_1} \mathbf{F}_{1_d}\, d\tau, \tag{19.1}$$

$$\int_{t_1}^{t_2} \left(\mathbf{F}_{1_r} + \mathbf{R}_1\right) d\tau \approx \int_{t_1}^{t_2} \mathbf{F}_{1_r}\, d\tau, \tag{19.2}$$

and similarly for body 2.

With equal and opposite collisional forces ($\mathbf{F}_{1_d} = -\mathbf{F}_{2_d}$, $\mathbf{F}_{1_r} = -\mathbf{F}_{2_r}$), we define the **coefficient of restitution**:

$$e = \frac{\int_{t_1}^{t_2} \mathbf{F}_{1_r} \cdot \mathbf{n}\, d\tau}{\int_{t_0}^{t_1} \mathbf{F}_{1_d} \cdot \mathbf{n}\, d\tau}. \tag{19.3}$$

Think!

Question: What does $e = 0$ and $e = 1$ mean?

Answer

- $e = 1$: perfectly **elastic** collision — compression impulse equals restitution impulse.
- $e = 0$: perfectly **plastic** collision — no restitution impulse.
- In general $0 \leq e \leq 1$, determined experimentally.

19.2.4 The Coefficient of Restitution in Terms of Velocities

From the linear impulse-momentum equations for each particle and stage:

$$\begin{aligned} m_1 v_{II} - m_1 \mathbf{v}_1 \cdot \mathbf{n} &= \int_{t_0}^{t_1} \mathbf{F}_{1_d} \cdot \mathbf{n} \, d\tau, \\ m_2 v_{II} - m_2 \mathbf{v}_2 \cdot \mathbf{n} &= \int_{t_0}^{t_1} \mathbf{F}_{2_d} \cdot \mathbf{n} \, d\tau, \\ m_1 \mathbf{v}_1' \cdot \mathbf{n} - m_1 v_{II} &= e \int_{t_0}^{t_1} \mathbf{F}_{1_d} \cdot \mathbf{n} \, d\tau, \\ m_2 \mathbf{v}_2' \cdot \mathbf{n} - m_2 v_{II} &= e \int_{t_0}^{t_1} \mathbf{F}_{2_d} \cdot \mathbf{n} \, d\tau. \end{aligned} \tag{19.4}$$

Eliminating v_{II} yields the familiar expression:

$$e = \frac{\mathbf{v}_2' \cdot \mathbf{n} - \mathbf{v}_1' \cdot \mathbf{n}}{\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}}. \tag{19.5}$$

! Note!

When one object impacts a massive constrained object (e.g. a wall), the velocity of the massive object is unaffected by the collision.

19.3 Example: The Most General Planar Impact Problem

Example

Question: Given particles m_1 and m_2 with pre-impact velocities $\mathbf{v}_1$ and $\mathbf{v}_2$, find the post-impact velocities. The coefficient of restitution e is given.

Answer

There are 4 unknowns (components of $\mathbf{v}_1'$ and $\mathbf{v}_2'$ along $\mathbf{n}$, $\mathbf{t}_1$, $\mathbf{t}_2$). Along the tangent directions: velocities are unchanged. Along $\mathbf{n}$: we have the conservation of linear momentum and the restitution equation.

19.4 Energy Loss During Impact

Pre-impact and post-impact kinetic energies:

$$T = \frac{1}{2}m_1\mathbf{v}_1 \cdot \mathbf{v}_1 + \frac{1}{2}m_2\mathbf{v}_2 \cdot \mathbf{v}_2, \tag{19.6}$$

$$T' = \frac{1}{2}m_1\mathbf{v}_1' \cdot \mathbf{v}_1' + \frac{1}{2}m_2\mathbf{v}_2' \cdot \mathbf{v}_2'. \tag{19.7}$$

The energy lost:

$$T - T' = \frac{m_1 m_2}{2(m_1 + m_2)} \left(\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}\right)^2 (1 - e^2). \tag{19.8}$$

Hence, for $0 \leq e \leq 1$, kinetic energy cannot increase as a result of impact.

19.5 Negative Values of the Coefficient of Restitution

Think!

Question: What does a negative e mean for a ball impacting a wall?

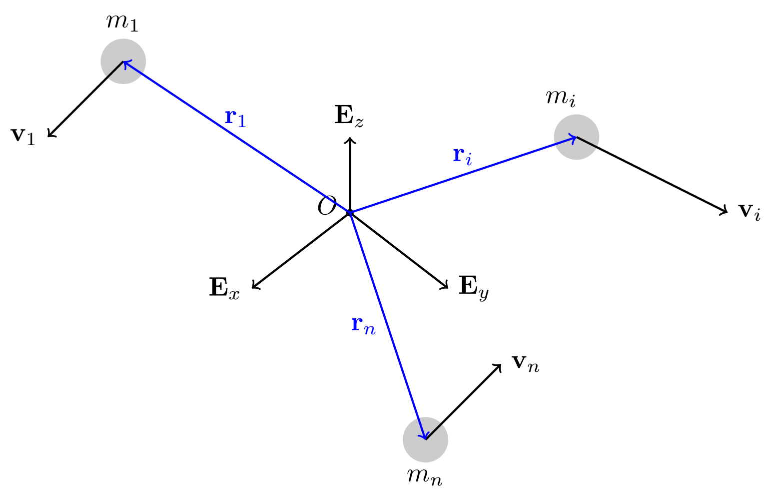
Answer

When $e < 0$, $\mathbf{v}_1' \cdot \mathbf{n} - \mathbf{v}_2' \cdot \mathbf{n}$ has the same sign as $\mathbf{v}_1 \cdot \mathbf{n} - \mathbf{v}_2 \cdot \mathbf{n}$, meaning the bodies pass through each other during impact. To prevent this, we generally require $e \geq 0$.

Part III

System of Particles

20 System of Particles



Consider a system of n particles. A typical particle has mass m_i, position vector $\mathbf{r}_i$ relative to O, velocity $\mathbf{v}_i = \dot{\mathbf{r}}_i$, acceleration $\mathbf{a}_i = \dot{\mathbf{v}}_i$, linear momentum $\mathbf{G}_i = m_i \mathbf{v}_i$, angular momentum $\mathbf{H}_i^P = (\mathbf{r}_i - \mathbf{r}_P) \times \mathbf{G}_i$, and kinetic energy $T_i = \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i$.

In this chapter, we define analogous system-level quantities and derive the balance laws for the system from the balance law for each individual particle.

20.1 Center of Mass

Think!

Question: What is the position vector of the center of mass of a system of particles?

Answer

The center of mass C has position vector

$$\mathbf{r}_C = \frac{1}{m}\sum_{k=1}^{n} m_k \mathbf{r}_k, \tag{20.1}$$

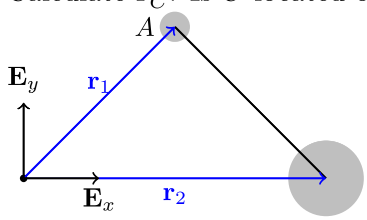
where $m = \sum_{k=1}^{n} m_k$ is the total mass.

Example

Question: Consider an unsymmetrical dumbbell: particle 1 of mass m and particle 2 of mass $3m$ connected by a massless rigid rod, with

$$\mathbf{r}_1 = \mathbf{E}_x + \mathbf{E}_y, \qquad \mathbf{r}_2 = 2\mathbf{E}_x.$$

Calculate $\mathbf{r}_C$. Is C located on the dumbbell?



Answer

$$\mathbf{r}_C = \frac{m\mathbf{r}_1 + 3m\mathbf{r}_2}{m+3m} = \frac{\mathbf{r}_1 + 3\mathbf{r}_2}{4} = \frac{7}{4}\mathbf{E}_x + \frac{1}{4}\mathbf{E}_y.$$

We can check that C lies on the line connecting m_1 and m_2.

Note the useful identity: relative to any reference point A,

$$\mathbf{r}_{C/A} = \frac{\sum_i m_i \mathbf{r}_{i/A}}{\sum_i m_i}. \tag{20.2}$$

Think!

Question: Show that $\sum_{k=1}^{n} m_k(\mathbf{r}_C - \mathbf{r}_k) = \mathbf{0}$ and $\sum_{k=1}^{n} m_k(\mathbf{v}_C - \mathbf{v}_k) = \mathbf{0}$.

Answer

Both follow directly from the definition of the center of mass.

Differentiating the position of the center of mass:

$$\mathbf{v}_C = \frac{1}{m}\sum_{k=1}^{n} m_k \mathbf{v}_k = \frac{1}{m}\sum_{k=1}^{n} \mathbf{G}_k. \tag{20.3}$$

20.2 Linear Momentum

The linear momentum of the system is

$$\mathbf{G} = \sum_{k=1}^{n} \mathbf{G}_k = m\mathbf{v}_C. \tag{20.4}$$

The linear momentum of the system equals that of its center of mass.

20.3 Angular Momentum

The angular momentum $\mathbf{H}^P$ of the system relative to a point P is:

$$\mathbf{H}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times m_k \mathbf{v}_k = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}, \tag{20.5}$$

where

$$\mathbf{H}^C = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_C) \times m_k \mathbf{v}_k = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_C) \times m_k(\mathbf{v}_k - \mathbf{v}_C). \tag{20.6}$$

Think!

Question: Show that $\sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times m_k \mathbf{v}_k = \mathbf{H}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{G}$.

Answer

Introduce $\mathbf{r}_C - \mathbf{r}_C$ inside the sum and expand.

20.4 Kinetic Energy

The kinetic energy of the system is:

$$T = \sum_{k=1}^{n} \frac{1}{2} m_k \mathbf{v}_k \cdot \mathbf{v}_k = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \sum_{k=1}^{n} m_k (\mathbf{v}_k - \mathbf{v}_C) \cdot (\mathbf{v}_k - \mathbf{v}_C). \tag{20.7}$$

The kinetic energy splits into the kinetic energy of the center of mass plus the kinetic energy relative to the center of mass. In general, $T \neq \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C$.

Think!

Question: Prove $\sum_{k=1}^{n} \frac{1}{2} m_k \mathbf{v}_k \cdot \mathbf{v}_k = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \sum_{k=1}^{n} m_k (\mathbf{v}_k - \mathbf{v}_C) \cdot (\mathbf{v}_k - \mathbf{v}_C)$.

Answer

Add $\mathbf{v}_C - \mathbf{v}_C$ to each $\mathbf{v}_k$, expand, and use $\sum m_k (\mathbf{v}_k - \mathbf{v}_C) = \mathbf{0}$.

20.5 Kinetics: Balance of Linear Momentum

Starting from $\mathbf{F}_i = m_i \mathbf{a}_i$ for each particle and summing:

$$\mathbf{F} = \sum_{k=1}^{n} \mathbf{F}_k = m \mathbf{a}_C. \tag{20.8}$$

The resultant external force equals the total mass times the acceleration of the center of mass.

Note!

The conditions for conservation of linear momentum of a system (completely or along a direction) are analogous to those for a single particle: $\mathbf{F} \cdot \mathbf{c} = 0$ for a constant $\mathbf{c}$.

20.6 Kinetics: Balance of Angular Momentum

Starting from $\mathbf{H}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times m_k \mathbf{v}_k$, the time derivative is:

$$\dot{\mathbf{H}}^P = \sum_{k=1}^{n} (\mathbf{r}_k - \mathbf{r}_P) \times \mathbf{F}_k - \mathbf{v}_P \times \mathbf{G} = \mathbf{M}^P - \mathbf{v}_P \times \mathbf{G}. \tag{20.9}$$

❗ Note!

$\dot{\mathbf{H}}^P$ is **not** necessarily equal to $\mathbf{M}^P$. The two agree only in special cases.

Two important special cases:

1. $P = O$ (fixed origin, $\mathbf{v}_O = \mathbf{0}$): $\dot{\mathbf{H}}^O = \mathbf{M}^O = \sum_{k=1}^n \mathbf{r}_k \times \mathbf{F}_k$.
2. $P = C$ (center of mass, $\mathbf{v}_C \times \mathbf{G} = \mathbf{0}$): $\dot{\mathbf{H}}^C = \mathbf{M}^C = \sum_{k=1}^n (\mathbf{r}_k - \mathbf{r}_C) \times \mathbf{F}_k$.

❗ Note!

The conditions for conservation of angular momentum of a system are analogous to those for a single particle.

Example: A pendulum suspended from a cart on smooth horizontal rails — write the BoAM in 3 forms.

20.7 Work-Energy Theorem

The work-energy theorem for a system of particles is:

$$\dot{T} = \sum_{k=1}^n \mathbf{F}_k \cdot \mathbf{v}_k, \tag{20.10}$$

and for energy conservation (with nonconservative forces $\mathbf{F}_{nc_k}$):

$$\dot{E} = \sum_{k=1}^n \mathbf{F}_{nc_k} \cdot \mathbf{v}_k. \tag{20.11}$$

Integrating:

$$T(t_2) - T(t_1) = \sum_{k=1}^n W_{\mathbf{F}_k,12}, \qquad W_{\mathbf{F}_k,12} = \int_{t_1}^{t_2} \mathbf{F}_k \cdot \mathbf{v}_k \, dt. \tag{20.12}$$

❗ Note!

The work of a force is the integral of the force dotted with the velocity of the **point of application** of the force.

Example

Question: Consider the unsymmetrical dumbbell (particle 1 of mass m, particle 2 of mass $3m$, massless rod) moving in the smooth $\{\mathbf{E}_x, \mathbf{E}_y\}$ plane with gravity along $-\mathbf{E}_z$. Is the energy conserved?

Answer

Drawing the FBDs of both masses: the only forces are gravity (conservative) and the constraint force in the rod (does no work on the system, since it is internal and the rod is rigid). Hence the total energy is conserved.

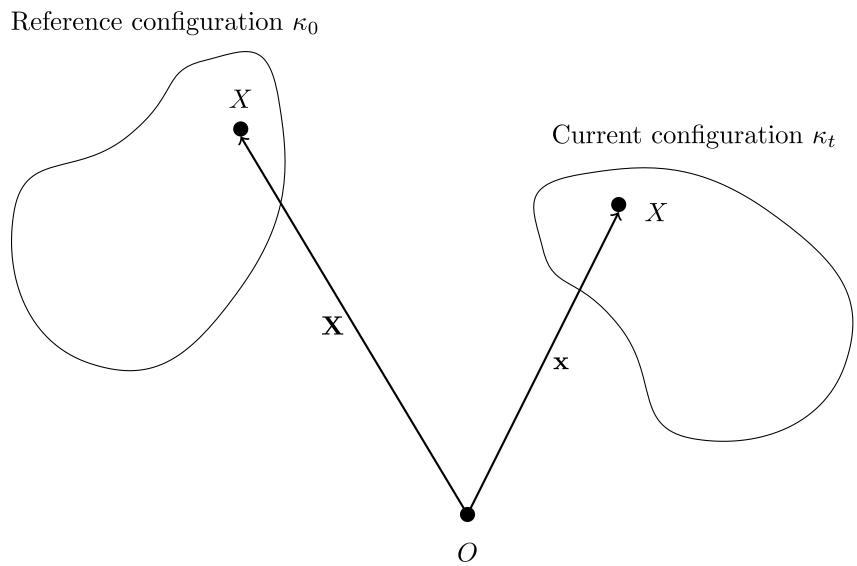
Part IV

Rigid Body Dynamics

21 Kinematics of Rigid Bodies

In this chapter, we will denote the fixed basis by $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$.

Consider a body $\mathcal{B}$ in motion in 3D space.



- A body $\mathcal{B}$ is a collection of material points (mass particles or particles).
- It is convenient to define a fixed reference configuration κ_0 for the body: $\mathbf{X} = \kappa_0$.
- The present (or current) configuration of the body is denoted by κ_t.
- We denote a material point by X. The position of the material point X relative to a fixed origin at the reference configuration is denoted by $\mathbf{X}$ and at time t is denoted by $\mathbf{x}$.
- This motion is invertible; we can use $\mathbf{X}$ to uniquely define the motion as a function of X and t: $\mathbf{x} = \chi(\mathbf{X}, t)$.
- χ (pronounced like "kye") is called the **motion function**. It tells you the current position vector of the material point which, at the reference configuration, had position vector $\mathbf{X}$.

Think!

Question: What is a mathematical statement that means that a body is rigid?

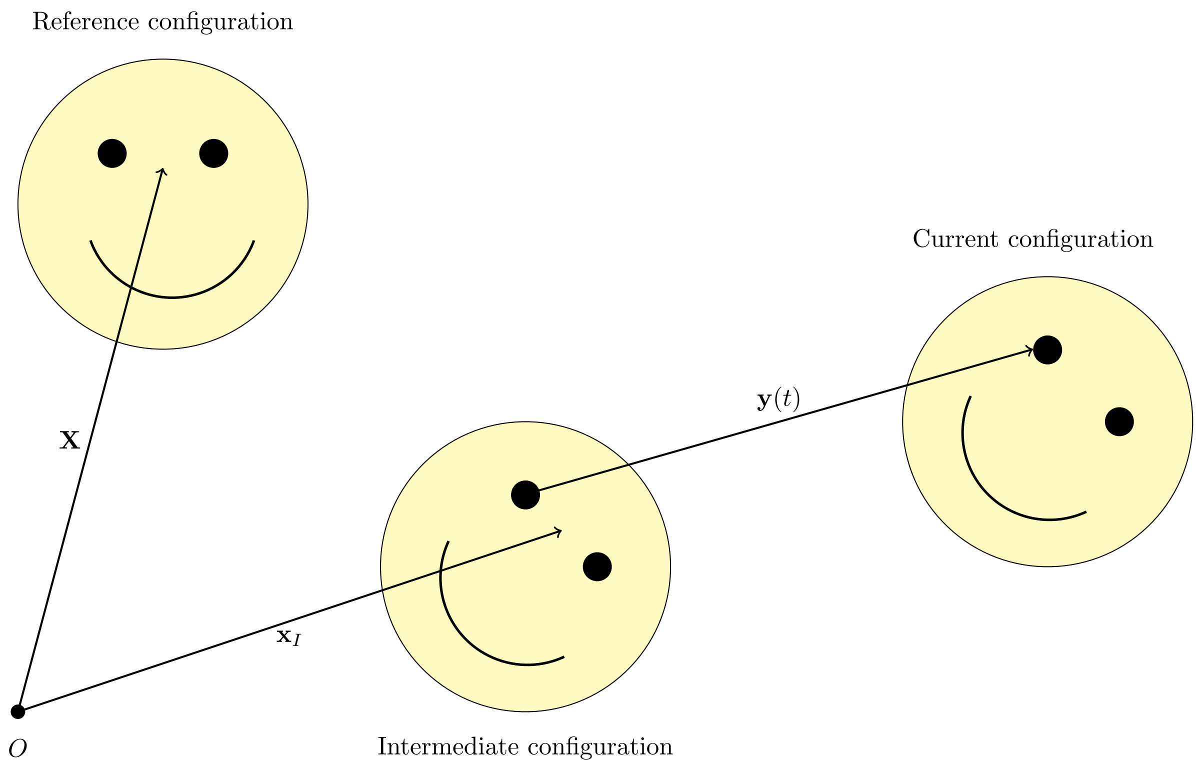
Answer

For rigid bodies the distance between any two mass particles X_1 and X_2 remains constant for all motions:

$$\|\mathbf{x}_1 - \mathbf{x}_2\| = \|\mathbf{X}_1 - \mathbf{X}_2\| \,. \tag{21.1}$$

The motion preserves angles between lines on the body (unlike elastic bodies or fluids) and preserves orientation (it is not a reflection).

Any rigid body motion can be built in two steps:



1. **Rotate** the object about O to the desired orientation: $\mathbf{x}_I = \mathbf{Q}(t)\mathbf{X}$ with $\|\mathbf{Q}(t)\mathbf{X}\| = \|\mathbf{X}\|$.
2. **Translate** via the translation vector $\mathbf{y}(t)$.

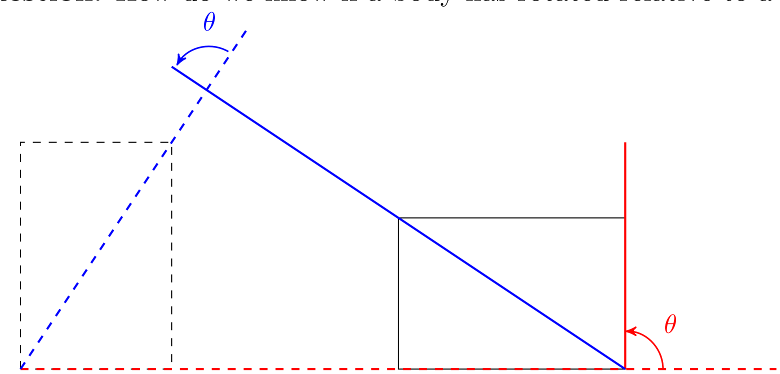
Then every rigid motion is:

$$\mathbf{x} = \chi(\mathbf{X}, t) = \mathbf{Q}(t)\mathbf{X} + \mathbf{y}(t). \tag{21.2}$$

$\mathbf{Q}(t)$ is a linear map and is called the **rotation tensor**. Both $\mathbf{Q}$ and $\mathbf{y}$ are the same for all material points (functions of t only). In this class we consider **general plane motion**, where $\omega = \dot{\theta}\mathbf{E}_3$ and $\alpha = \dot{\omega}$.

Think!

Question: How do we know if a body has rotated relative to a reference configuration?



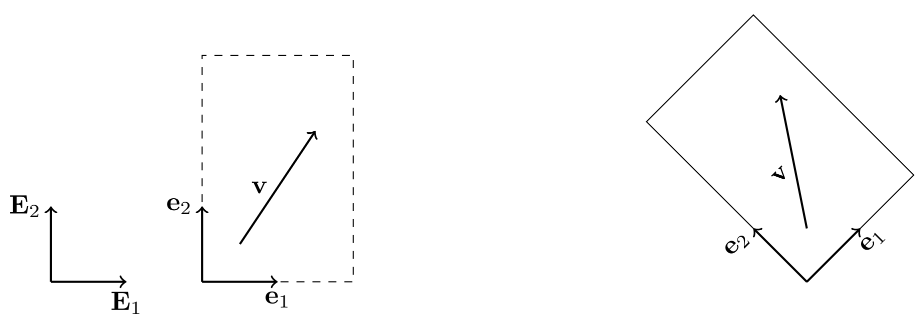
reference configuration current configuration

Answer

We need the reference configuration to define rotation. The rotation angle θ is the angular displacement of every material line from reference to current configuration. The axis of rotation is perpendicular to the page.

21.1 Corotational Bases

A *corotational basis* $\{\mathbf{e}_1, \mathbf{e}_2\}$ is a pair of vectors embedded in the material. Initially $\mathbf{e}_i = \mathbf{E}_i$.



reference configuration current configuration

For a rigid body, the components of a material vector $\mathbf{v}$ along the corotational basis are constant throughout the motion:

$$\mathbf{v} = a\mathbf{e}_1 + b\mathbf{e}_2, \tag{21.3}$$

where a and b are constant and $\mathbf{e}_1(t)$, $\mathbf{e}_2(t)$ evolve with the body. The corotational basis is unaffected by translation.

21.2 Tensors

A tensor $\mathbf{T}$ is a *linear transformation* that transforms a vector into another vector: $\mathbf{Tu} = \mathbf{v}$. A tensor exists independent of any basis and is defined by its action on vectors.

Examples: a rotation tensor $\mathbf{Q}$, the identity tensor $\mathbf{I}$, a scaling tensor $\mathbf{S}$.

Think!

Question: Is a rotation a linear transformation?

Answer

Yes. Rotating $a\mathbf{X}$ is the same as rotating $\mathbf{X}$ then scaling by a, and rotating $\mathbf{X}_1 + \mathbf{X}_2$ equals rotating each and adding. So rotations are linear transformations — hence tensors.

Think!

Question: Are vectors and matrices the same?

Answer

No. A vector is an abstract object independent of any basis; a matrix is its representation in a specific basis. Similarly, a tensor is not a matrix, but can be expressed as one in a given coordinate system.

21.3 Tensor (Bun) Product

The **tensor product $\mathbf{a} \otimes \mathbf{b}$** acts on vectors by:

$$(\mathbf{a} \otimes \mathbf{b})\, \mathbf{u} = (\mathbf{b} \cdot \mathbf{u})\, \mathbf{a}. \tag{21.4}$$

This is linear (verifiable directly), so it is a tensor.

Think!

Question: What does $2\mathbf{E}_1 \otimes \mathbf{E}_1 + 2\mathbf{E}_2 \otimes \mathbf{E}_2 + 2\mathbf{E}_3 \otimes \mathbf{E}_3$ do to a vector?

Answer

It scales it by a factor of two.

Think!

Question: Show that $\mathbf{E}_1 \otimes \mathbf{E}_1 + \mathbf{E}_2 \otimes \mathbf{E}_2 + \mathbf{E}_3 \otimes \mathbf{E}_3$ is the identity tensor. Show also that the rotation tensor can be expressed as $\mathbf{Q} = \mathbf{e}_x \otimes \mathbf{E}_x + \mathbf{e}_y \otimes \mathbf{E}_y + \mathbf{e}_z \otimes \mathbf{E}_z$.

Answer

Apply the definition of the tensor product term by term.

21.4 Matrix Representation of Tensors

Given a basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, define $Q_{ij} \equiv \left(\mathbf{Q}\mathbf{e}_j\right) \cdot \mathbf{e}_i$. Then the action of $\mathbf{Q}$ in matrix form is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}. \tag{21.5}$$

The matrix representation depends on the chosen basis, but the tensor itself is basis-independent.

21.4.1 Rigid Body Motion

In matrix form, rigid body motion is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} Q_{11}(t) & Q_{12}(t) & Q_{13}(t) \\ Q_{21}(t) & Q_{22}(t) & Q_{23}(t) \\ Q_{31}(t) & Q_{32}(t) & Q_{33}(t) \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}. \tag{21.6}$$

For $[\mathbf{Q}(t)]$ to be a rotation matrix: $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ and $\det([\mathbf{Q}]) = 1$.

Think!

Question: Verify that the rotation about $\mathbf{E}_z$ by angle $\theta(t)$ has matrix representation

$$\mathtt{Q} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

using $\mathbf{Q} = \mathbf{e}_1 \otimes \mathbf{E}_1 + \mathbf{e}_2 \otimes \mathbf{E}_2 + \mathbf{e}_3 \otimes \mathbf{E}_3$ with $\mathbf{e}_1 = \cos\theta\,\mathbf{E}_1 + \sin\theta\,\mathbf{E}_2$ and $\mathbf{e}_2 = -\sin\theta\,\mathbf{E}_1 + \cos\theta\,\mathbf{E}_2$.

Answer

Compute $Q_{ij} = (\mathbf{QE}_j) \cdot \mathbf{E}_i = \mathbf{e}_j \cdot \mathbf{E}_i$ to recover the matrix.

! Note!

This matrix satisfies $\mathtt{QQ}^T = \mathtt{I}$ and $\det(\mathtt{Q}) = 1$, so $\mathbf{Q}$ is a **proper orthogonal tensor**. All proper orthogonal tensors are rotations and vice versa.

21.5 The Angular Velocity Vector

From rigid body motion, for any two material points A and B: $(\mathbf{x}_B - \mathbf{x}_A) = \mathbf{Q}(\mathbf{X}_B - \mathbf{X}_A)$. Differentiating:

$$\mathbf{v}_B - \mathbf{v}_A = \dot{\mathbf{Q}}\mathbf{Q}^T(\mathbf{x}_B - \mathbf{x}_A). \tag{21.7}$$

Since $\mathbf{QQ}^T = \mathbf{I}$, $\dot{\mathbf{Q}}\mathbf{Q}^T$ is **skew**. Every skew tensor corresponds to a vector cross product, so:

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{x}_B - \mathbf{x}_A). \tag{21.8}$$

The tensor $\Omega = \dot{\mathbf{Q}}\mathbf{Q}^T$ is the **angular velocity tensor** and ω is the **angular velocity vector**. Taking another derivative:

$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) \tag{21.9}$$

where α is the angular acceleration.

! Note!

For any corotational basis vector $\mathbf{e}_i$ fixed to the rigid body:

$$\dot{\mathbf{e}}_i = \omega \times \mathbf{e}_i, \quad i = 1, 2, 3. \tag{21.10}$$

21.6 Velocity and Acceleration of Two Material Points

Let A and B be material points on the rigid body. Expressing $\mathbf{x}_{B/A}$ on a corotational basis (constant components):

$$\mathbf{x}_{B/A} = x\mathbf{e}_x + y\mathbf{e}_y, \tag{21.11}$$

$$\mathbf{v}_{B/A} = x\dot{\mathbf{e}}_x + y\dot{\mathbf{e}}_y = \omega \times \mathbf{x}_{B/A}, \tag{21.12}$$

$$\mathbf{a}_{B/A} = \alpha \times \mathbf{x}_{B/A} + \omega \times (\omega \times \mathbf{x}_{B/A}). \tag{21.13}$$

With $\omega = \omega \mathbf{E}_z$ and $\alpha = \alpha \mathbf{E}_z$:

$$\omega\, d\omega = \alpha\, d\theta, \qquad \omega\, dt = d\theta, \qquad \alpha\, dt = d\omega. \tag{21.14}$$

21.7 Particle in a Noninertial Frame

A noninertial frame is one that accelerates (e.g. an accelerating bus, a merry-go-round). If A is fixed to the rigid body and B moves relative to it:

$$\mathbf{v}_B - \mathbf{v}_A = \omega \times (\mathbf{x}_B - \mathbf{x}_A) + \mathring{\mathbf{x}}_{B/A}, \tag{21.15}$$

$$\mathbf{a}_B - \mathbf{a}_A = \alpha \times (\mathbf{x}_B - \mathbf{x}_A) + \omega \times (\omega \times (\mathbf{x}_B - \mathbf{x}_A)) + 2\omega \times \mathring{\mathbf{x}}_{B/A} + \overset{\circ\circ}{\mathbf{x}}_{B/A}, \tag{21.16}$$

where $\mathring{\mathbf{x}}_{B/A} = \dot{x}\mathbf{e}_x + \dot{y}\mathbf{e}_y$ and $\overset{\circ\circ}{\mathbf{x}}_{B/A} = \ddot{x}\mathbf{e}_x + \ddot{y}\mathbf{e}_y$ are the first and second **corotational rates**. The term $2\omega \times \mathring{\mathbf{x}}_{B/A}$ is the **Coriolis acceleration** and $\omega \times (\omega \times \mathbf{x}_{B/A})$ is the **centripetal acceleration**.

Think!

Question: See this video of people playing catch on a merry-go-round. A ball is thrown on a merry-go-round rotating about fixed center O, with $\mathbf{r}_0 = -R\mathbf{E}_x$ and $\mathbf{v}_0 = v_0\mathbf{E}_x$. What velocity does the ball appear to have to an observer on the merry-go-round?

Answer

The ball is a projectile: $\mathbf{a} = -g\mathbf{E}_z$, $\mathbf{v} = -gt\mathbf{E}_z + \mathbf{v}_0$. From $\mathbf{v} = \omega \times (\mathbf{r} - \mathbf{r}_0) + \mathring{\mathbf{x}}_{P/O}$:

$$\mathring{\mathbf{x}}_{P/O} = -gt\mathbf{E}_z + v_0\mathbf{E}_x + \omega(v_0 t - R)\mathbf{E}_y. \tag{21.17}$$

21.8 Classifications of Rigid Body Motions

Pure (rectilinear) translation: $\omega = \mathbf{0}$, $\alpha = \mathbf{0}$.

Curvilinear translation: material points follow parallel curved paths; $\omega = \mathbf{0}$, $\alpha = \mathbf{0}$ (e.g. Ferris wheel bucket).

Fixed point rotation: $\omega \neq \mathbf{0}$, fixed point with $\mathbf{v}_O = \mathbf{a}_O = \mathbf{0}$ (e.g. pendulum):

$$\mathbf{a}_P = \alpha \times \mathbf{r}_P + \omega \times (\omega \times \mathbf{r}_P). \tag{21.18}$$

The term $\alpha \times \mathbf{r}_P$ is the tangential acceleration; $\omega \times (\omega \times \mathbf{r}_P)$ is the centripetal acceleration.

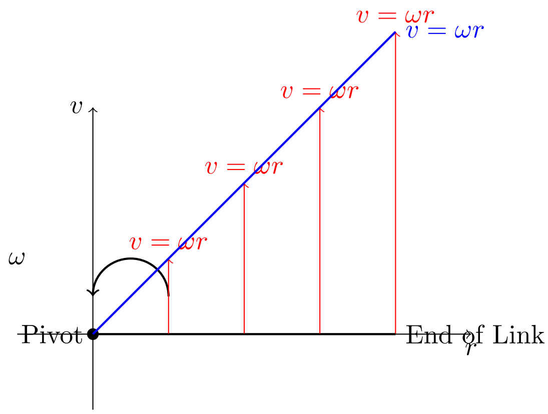
General motion: combined translation and rotation.

21.9 Instantaneous Center of Rotation

Think!

Question: Draw the velocity profile of a rotating link with a fixed end.

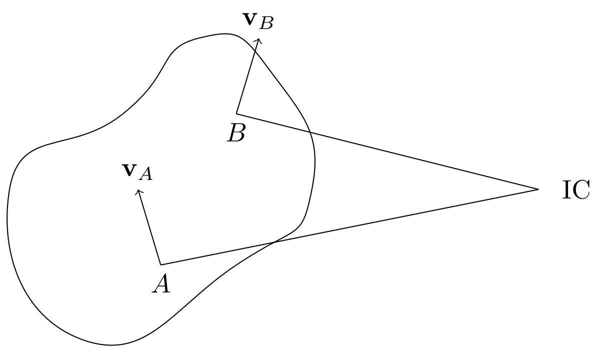
Answer



A body in general plane motion is not rotating about a fixed point, but **instantaneously** it rotates about the **instantaneous center of rotation (IC)** — a point with zero velocity at that instant. If $\mathbf{v}_{IC} = \mathbf{0}$, then for any material point A:

$$\mathbf{v}_A = \omega \times (\mathbf{r}_A - \mathbf{r}_{IC}). \tag{21.19}$$

So $\mathbf{r}_{A/IC}$ is perpendicular to $\mathbf{v}_A$. Knowing the directions of $\mathbf{v}_A$ and $\mathbf{v}_B$ for two material points identifies the IC.



For the IC:

$$\mathbf{v}_P = \omega \times \mathbf{r}_{P/IC}, \qquad \mathbf{a}_P = \alpha \times \mathbf{r}_{P/IC} + \omega \times (\omega \times \mathbf{r}_{P/IC}). \tag{21.20}$$

Remarks: the IC generally migrates through space; for a translating rigid body the IC is at infinity; the IC can only be used for velocity analysis since its acceleration is nonzero.

Procedure: use geometry to find the IC; find $|\omega|$ via $\mathbf{v}_P = \omega \times \mathbf{r}_{P/IC}$; obtain $\mathbf{v}_P$ for any point.

Examples: Crank slider, rod falling on a corner.

21.10 Doing Kinematics Geometrically

In some problems, geometric laws that hold for all time can be differentiated: - The law of sines holds at all times. - The law of cosines holds at all times. - Angle sum relations for a triangle (e.g. $\beta + \alpha + \theta = \pi$) can be differentiated.

21.11 Kinematics of Rolling and Sliding

> **!** Note!
>
> Terminology: "rolling" = roll without slip; "sliding" = roll with slip.

Consider a rigid body $\mathcal{B}$ in contact with a fixed surface $\mathcal{S}$. Let P be the material point of $\mathcal{B}$ in contact at time t, with unit normal $\mathbf{n}$ to $\mathcal{S}$.

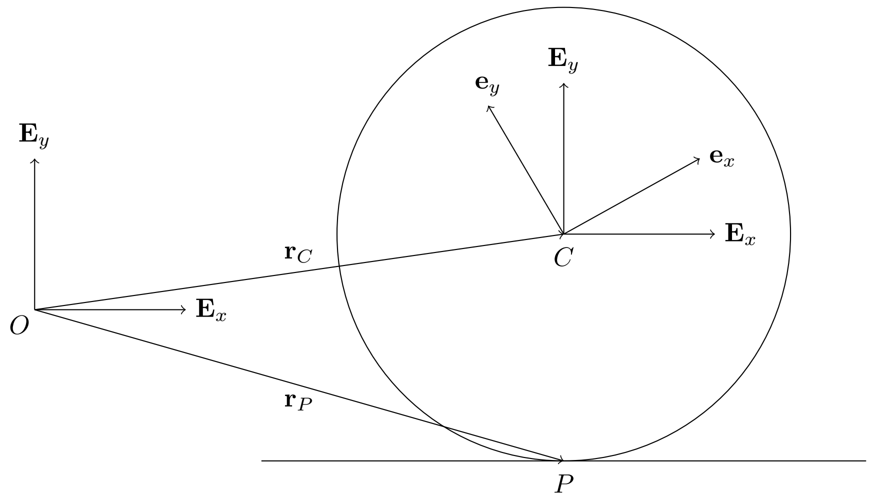
Since P belongs to $\mathcal{B}$:

$$\mathbf{v}_P - \mathbf{v}_C = \omega \times (\mathbf{r}_P - \mathbf{r}_C), \tag{21.21}$$
$$\mathbf{a}_P - \mathbf{a}_C = \alpha \times (\mathbf{r}_P - \mathbf{r}_C) + \omega \times (\omega \times (\mathbf{r}_P - \mathbf{r}_C)) \,. \tag{21.22}$$

- **Contact (touching):** $\mathbf{v}_P \cdot \mathbf{n} = 0$.
- **Rolling (no slip):** $\mathbf{v}_P = \mathbf{0}$, so $\mathbf{v}_C = -\omega \times (\mathbf{r}_P - \mathbf{r}_C)$. Note: $\mathbf{a}_P \neq \mathbf{0}$ in general.

21.11.1 Example: Rolling Circular Disk

> Think!
>
> **Question:** What is the IC of a rolling disk?
>
> **Answer**
>
> The contact point with the surface.



A disk of radius R rolling on a plane with $\omega = \omega \mathbf{E}_z$. Since $\mathbf{v}_P = \mathbf{0}$:

$$\mathbf{v}_C = \omega \times R\mathbf{E}_y = -\omega R \mathbf{E}_x \implies \dot{x} = -\omega R, \quad \ddot{x} = -\alpha R. \tag{21.23}$$

The acceleration of the contact point:

$$\mathbf{a}_P = -\omega^2 R \mathbf{E}_y \tag{21.24}$$

(purely vertical — toward the center).

21.12 Center of Mass

The center of mass C of $\mathcal{B}$ has position vector

$$\mathbf{r}_C = \frac{\int_{\mathcal{B}} \mathbf{r}\, dm}{\int_{\mathcal{B}} dm}, \qquad dm = \rho(x,t)\, dv. \tag{21.25}$$

For rigid bodies, C behaves as a material point:

$$\mathbf{v}_C - \mathbf{v}_A = \omega \times (\mathbf{r}_C - \mathbf{r}_A), \qquad \mathbf{a}_C - \mathbf{a}_A = \alpha \times (\mathbf{r}_C - \mathbf{r}_A) + \omega \times (\omega \times (\mathbf{r}_C - \mathbf{r}_A)). \tag{21.26}$$

21.13 Linear Momentum

$$\mathbf{G} = \int_{\mathcal{B}} \mathbf{v}\, dm = m\mathbf{v}_C. \tag{21.27}$$

21.14 Angular Momenta

The angular momentum about a point P is

$$\mathbf{H}^P = \int_{\mathcal{B}} (\mathbf{r} - \mathbf{r}_P) \times \mathbf{v}\, dm, \tag{21.28}$$

and the relationship between angular momenta about O and C is

$$\mathbf{H}^O = \mathbf{H}^C + \mathbf{r}_C \times \mathbf{G}. \tag{21.29}$$

21.14.1 Inertia Tensor

With $\pi = \mathbf{r} - \mathbf{r}_C$ and using the BAC-CAB identity:

$$\mathbf{H}^C = \int_{\mathcal{B}} ((\pi \cdot \pi)\omega - (\pi \cdot \omega)\pi)\, dm = \mathbf{I}^C \omega. \tag{21.30}$$

The inertia matrix on the corotational basis:

$$[\mathbf{I}^C] = \begin{bmatrix} I^C_{xx} & I^C_{xy} & I^C_{xz} \\ I^C_{xy} & I^C_{yy} & I^C_{yz} \\ I^C_{xz} & I^C_{yz} & I^C_{zz} \end{bmatrix}, \tag{21.31}$$

with diagonal entries (**moments of inertia**) and off-diagonal entries (**products of inertia**):

$$I^C_{xx} = \int_{\mathcal{B}} (y^2 + z^2)\, dm, \qquad I^C_{yy} = \int_{\mathcal{B}} (x^2 + z^2)\, dm, \qquad I^C_{zz} = \int_{\mathcal{B}} (x^2 + y^2)\, dm, \tag{21.32}$$

$$I^C_{xy} = -\int_{\mathcal{B}} xy\, dm, \qquad I^C_{xz} = -\int_{\mathcal{B}} xz\, dm, \qquad I^C_{yz} = -\int_{\mathcal{B}} yz\, dm. \tag{21.33}$$

When $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ is an eigenbasis of $[\mathbf{I}^C]$, the products of inertia vanish.

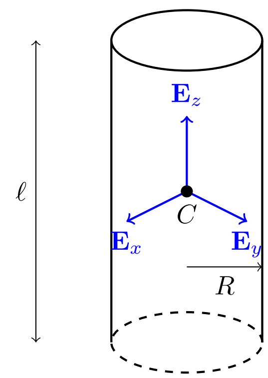
Think!

Question: Is my moment of inertia about an axis through my body greater when my arms are closed or open?

Answer

Greater when arms are open — mass is farther from the axis. See also: Professor Walter Lewin rolling cylinders.

21.14.2 Example: Solid Cylinder



With $dm = \rho\, r\, dr\, d\theta\, dz$:

$$I_{zz}^C = \rho \int_{-\ell/2}^{\ell/2} \int_0^{2\pi} \int_0^R r^3\, dr\, d\theta\, dz = \frac{1}{2}\rho\pi\ell R^4 = \frac{mR^2}{2}. \tag{21.34}$$

21.14.3 Example: Cylindrical Hoop

Every dm has $r = R$, so:

$$I_{zz}^C = R^2 \int_{\mathcal{B}} dm = mR^2. \tag{21.35}$$

21.15 Parallel Axis Theorem

Given inertia components about C, the inertia about a material point A with $\mathbf{r}_A - \mathbf{r}_C = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z$ (axes parallel at A and C):

$$I_{xx}^A = I_{xx}^C + m(A_y^2 + A_z^2), \tag{21.36}$$

$$I_{xy}^A = I_{xy}^C - mA_xA_y. \tag{21.37}$$

In tensor form:

$$\mathbf{I}^A = \mathbf{I}^C + m\,\|\mathbf{r}_A - \mathbf{r}_C\|^2\,\mathbf{I} - m(\mathbf{r}_A - \mathbf{r}_C) \otimes (\mathbf{r}_A - \mathbf{r}_C). \tag{21.38}$$

Note!

The parallel axis theorem applies only between the center of mass C and another material point — not between any two arbitrary material points. The moment of inertia always increases when moving away from C.

21.16 Radius of Gyration

The radius of gyration k_z about the z-axis satisfies $mk_z^2 = I_{zz}$.

21.17 Angular Momentum About a Moving Point P

For a material point P on the rigid body:

$$\mathbf{H}^P = \mathbf{I}^P \omega + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P, \tag{21.39}$$

with special cases $\mathbf{H}^C = \mathbf{I}^C \omega$ and $\mathbf{H}^O = \mathbf{I}^O \omega$ (fixed O). In general:

$$\dot{\mathbf{H}}^P = \mathbf{I}^P \mathring{\omega} + \omega \times (\mathbf{I}^P \omega) + \frac{d}{dt}\left[(\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{v}_P\right]. \tag{21.40}$$

22 Kinetics of Rigid Bodies

22.1 Balance Laws for a Rigid Body

The balance laws for a rigid body are **Euler's first law** (Balance of Linear Momentum, BoLM) and **Euler's second law** (Balance of Angular Momentum, BoAM):

$$\mathbf{F} = m\mathbf{a}_C, \tag{22.1}$$

$$\mathbf{M}^O = \dot{\mathbf{H}}^O \quad \text{about a fixed point } O, \tag{22.2}$$

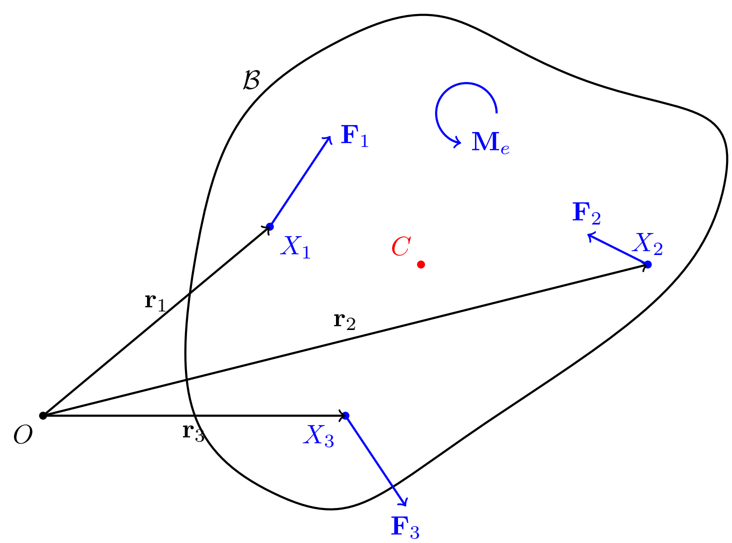
$$\mathbf{M}^C = \dot{\mathbf{H}}^C \quad \text{about the center of mass } C, \tag{22.3}$$

$$\mathbf{M}^P = \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G} \quad \text{about any material point } P. \tag{22.4}$$

The BoAM is an **independent postulate**, not derivable from the BoLM.

22.2 Resultant Forces and Moments

The resultant force $\mathbf{F}$ is the sum of all external forces. The resultant moment about a point includes the moment due to forces plus any applied pure moments $\mathbf{M}_e$ not due to forces.



For K forces $\mathbf{F}_i$ acting at positions $\mathbf{r}_i$ plus a pure moment $\mathbf{M}_e$:

$$\mathbf{F} = \sum_{i=1}^{K} \mathbf{F}_i, \tag{22.5}$$

$$\mathbf{M}^O = \mathbf{M}_e + \sum_{i=1}^{K} \mathbf{r}_i \times \mathbf{F}_i, \tag{22.6}$$

$$\mathbf{M}^C = \mathbf{M}_e + \sum_{i=1}^{K} (\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i. \tag{22.7}$$

Examples of pure moments: reaction moments $\mathbf{M}_R$ at joints; torsional spring moments $-K_T(\theta - \theta_0)\mathbf{E}_z$.

22.2.1 Does Weight Give a Moment About the Center of Mass?

The total weight is $\mathbf{W} = \int_{\mathcal{B}} \mathbf{g}\, dm = m\mathbf{g}$, and its moment about C is:

$$\mathbf{M}^C = \int_{\mathcal{B}} \pi \times \mathbf{g}\, dm = \left(\int_{\mathcal{B}} \pi\, dm \right) \times \mathbf{g} = \mathbf{0}, \tag{22.8}$$

since $\int_{\mathcal{B}} \pi\, dm = \mathbf{0}$ by definition of the center of mass.

22.3 Equivalence of the BoAM Forms

22.3.1 BoAM About Any Point P

Starting from $\mathbf{M}^C = \dot{\mathbf{H}}^C$ and replacing $\mathbf{M}^P = \mathbf{M}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{F}$:

$$\mathbf{M}^P = \dot{\mathbf{H}}^C + (\mathbf{r}_C - \mathbf{r}_P) \times \mathbf{F}. \tag{22.9}$$

Substituting $\dot{\mathbf{H}}^C = \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G} + (\mathbf{r}_P - \mathbf{r}_C) \times m\mathbf{a}_C$ and simplifying:

$$\mathbf{M}^P = \dot{\mathbf{H}}^P + (\mathbf{v}_P - \mathbf{v}_C) \times \mathbf{G}, \tag{22.10}$$

recovering the BoAM about a general material point P.

22.3.2 BoAM About a Fixed Point O

If P is a fixed point O (so $\mathbf{v}_O = \mathbf{0}$), the above simplifies to $\mathbf{M}^O = \dot{\mathbf{H}}^O$.

22.4 Calculating the Derivative of Angular Momentum

Recall $\mathbf{H}^C = \mathbf{I}^C\omega$ and $\dot{\mathbf{H}}^C = \overset{\circ}{\mathbf{H}}^C + \omega \times \mathbf{H}^C$, where $\overset{\circ}{\mathbf{H}}^C$ is the corotational rate (time derivative with $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ fixed).

For $\omega = \omega\mathbf{E}_z$ (the case exclusively considered in this class):

$$\mathbf{H}^C = I^C_{xz}\omega_z\mathbf{e}_x + I^C_{yz}\omega_z\mathbf{e}_y + I^C_{zz}\omega_z\mathbf{e}_z, \tag{22.11}$$

$$\dot{\mathbf{H}}^C = (I^C_{xz}\dot{\omega} - I^C_{yz}\omega^2)\mathbf{e}_x + (I^C_{yz}\dot{\omega} + I^C_{xz}\omega^2)\mathbf{e}_y + I^C_{zz}\dot{\omega}\mathbf{E}_z. \tag{22.12}$$

Analogously for other reference points.

! Note!

For fixed axis rotation, sum moments about a fixed point on the axis. For general plane motion, sum about C or a general material point P using

$$\mathbf{M}^P = (I^C_{xz}\dot{\omega} - I^C_{yz}\omega^2)\mathbf{e}_x + (I^C_{yz}\dot{\omega} + I^C_{xz}\omega^2)\mathbf{e}_y + I^C_{zz}\dot{\omega}\mathbf{E}_z + (\mathbf{r}_C - \mathbf{r}_P) \times m\mathbf{a}_C. \tag{22.13}$$

In most problems, axes are chosen so that $I_{xz} = I_{yz} = 0$.

22.5 Conservative Moments

22.5.1 Torsional Spring

A torsional spring with stiffness K creates a couple $\mathbf{M}_S = -K\theta\mathbf{E}_z$. Its power is:

$$\mathbf{M}_S \cdot \omega = -K\theta\dot{\theta} = -\frac{d}{dt}\left(\frac{K\theta^2}{2}\right). \tag{22.14}$$

Hence $U = \frac{1}{2}K\theta^2$ for the torsional spring.

22.5.2 Constant Couple

A constant couple $\mathbf{M}_e = M_e\mathbf{E}_z$ has power $\mathbf{M}_e \cdot \omega = M_e\dot{\theta} = -\frac{d}{dt}(-M_e\theta)$. Hence $U = -M_e\theta$.

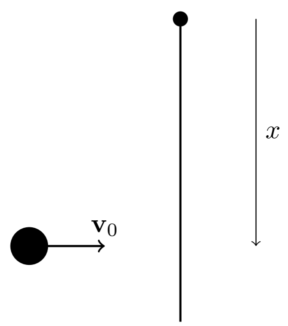
! Note!

These results apply only for $\omega = \omega\mathbf{E}_z$. Constant couples are not conservative in general.

22.6 Impact Problems and Center of Percussion

For rigid bodies, impact ideas are the same as in systems of particles.

Example: Consider a projectile fired horizontally at a pendulum (in the horizontal plane, no gravity). Find position x along the pendulum such that the reaction force at the pin is nearly zero during impact.



If x is large, the horizontal reaction at O points left; if x is small, it points right. The optimal x that minimizes the reaction is called the **center of percussion**.

22.7 Summary

For K forces and a moment $\mathbf{M}_e$ acting on a rigid body, the balance laws are:

$$\mathbf{F} = m\frac{d\mathbf{v}_C}{dt}, \qquad \mathbf{M}^O = \dot{\mathbf{H}}^O \quad \text{(or equivalently } \mathbf{M}^C = \dot{\mathbf{H}}^C\text{)}. \tag{22.15}$$

For fixed-axis rotation with $\omega = \omega\mathbf{E}_z$ and $I_{xz} = I_{yz} = 0$:

$$\mathbf{F} = m\dot{\mathbf{v}}_C, \qquad \mathbf{M} = I_{zz}\dot{\omega}\mathbf{E}_z. \tag{22.16}$$

Four classes of applications: (1) purely translational motion ($\omega = \alpha = \mathbf{0}$), (2) rigid body with a fixed point, (3) rolling and sliding rigid bodies, (4) imbalanced rotors.

23 The Work-Energy Theorem for a Rigid Body

The **Koenig decomposition** for the kinetic energy of a rigid body is:

$$T = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \mathbf{H}^C \cdot \omega. \tag{23.1}$$

In this class, $\omega = \omega \mathbf{E}_z$, so $\frac{1}{2}\mathbf{H}^C \cdot \omega = \frac{1}{2} I_{zz} \omega^2$.

For fixed point rotation about O, the kinetic energy simplifies to:

$$T = \frac{1}{2} \mathbf{H}^O \cdot \omega = \frac{1}{2} I_{zz}^O \omega^2. \tag{23.2}$$

The **work-energy theorem** for a rigid body is:

$$\frac{dT}{dt} = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega = \sum_{i=1}^{N} \mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_e \cdot \omega. \tag{23.3}$$

Integrating from t_A to t_B:

$$T_B - T_A = W_{\mathbf{F},AB} + W_{\mathbf{M},AB} = \sum_{i=1}^{N} W_{\mathbf{F}_i,AB} + W_{\mathbf{M}_e,AB}, \tag{23.4}$$

where

$$W_{\mathbf{F},AB} = \int_{t_A}^{t_B} \mathbf{F} \cdot \mathbf{v}_C \, dt, \qquad W_{\mathbf{M},AB} = \int_{t_A}^{t_B} \mathbf{M}^C \cdot \omega \, dt, \tag{23.5}$$

$$W_{\mathbf{F}_i,AB} = \int_{t_A}^{t_B} \mathbf{F}_i \cdot \mathbf{v}_i \, dt, \qquad W_{\mathbf{M}_e,AB} = \int_{t_A}^{t_B} \mathbf{M}_e \cdot \omega \, dt. \tag{23.6}$$

Notice: the work of a force uses the velocity of its **point of application**.

23.1 Koenig's Decomposition

By definition, the kinetic energy of a rigid body is:

$$T = \frac{1}{2} \int_{\mathcal{B}} \mathbf{v} \cdot \mathbf{v} \, \rho \, dv. \tag{23.7}$$

Using $\mathbf{v} = \mathbf{v}_C + \omega \times \pi$ with $\pi = \mathbf{r} - \mathbf{r}_C$:

$$T = \frac{1}{2} \int_{\mathcal{B}} (\mathbf{v}_C \cdot \mathbf{v}_C + 2\mathbf{v}_C \cdot (\omega \times \pi) + (\omega \times \pi) \cdot (\omega \times \pi))\, dm. \tag{23.8}$$

The first term integrates to $\frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C$, the second vanishes (since $\int \pi\, dm = \mathbf{0}$), and using the BAC-CAB identity for the third:

$$T = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \mathbf{H}^C \cdot \omega. \tag{23.9}$$

23.1.1 Fixed Point Rotation

For fixed point rotation about O ($\mathbf{v}_O = \mathbf{0}$), using the cyclic property of the scalar triple product:

$$T = \frac{1}{2} \mathbf{G} \cdot (\omega \times \mathbf{r}_{C/O}) + \frac{1}{2} \mathbf{H}^C \cdot \omega = \frac{1}{2} (\mathbf{r}_{C/O} \times \mathbf{G} + \mathbf{H}^C) \cdot \omega = \frac{1}{2} \mathbf{H}^O \cdot \omega. \tag{23.10}$$

23.2 Derivation of the Work-Energy Theorem

From $T = \frac{1}{2} m \mathbf{v}_C \cdot \mathbf{v}_C + \frac{1}{2} \mathbf{H}^C \cdot \omega$, taking the time derivative and using the fact that $\dot{\mathbf{H}}^C \cdot \omega = \mathbf{H}^C \cdot \dot{\omega}$ (verified by direct calculation):

$$\dot{T} = m \dot{\mathbf{v}}_C \cdot \mathbf{v}_C + \dot{\mathbf{H}}^C \cdot \omega. \tag{23.11}$$

Invoking the BoLM and BoAM:

$$\dot{T} = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}^C \cdot \omega. \tag{23.12}$$

23.3 Alternative Form

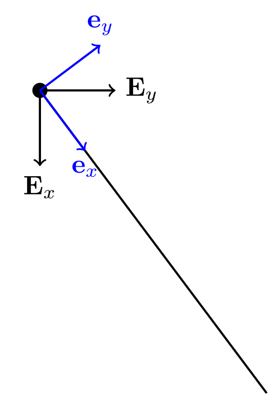
Using $\mathbf{F} = \sum_i \mathbf{F}_i$ and $\mathbf{M}^C = \mathbf{M}_e + \sum_i (\mathbf{r}_i - \mathbf{r}_C) \times \mathbf{F}_i$, and noting that $\mathbf{v}_i = \mathbf{v}_C + \omega \times (\mathbf{r}_i - \mathbf{r}_C)$:

$$\dot{T} = \sum_{i=1}^{K} \mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_e \cdot \omega. \tag{23.13}$$

This form is often easier to use in applications: each force is dotted with the velocity of its point of application.

23.4 Examples

23.4.1 Bar Pendulum Fixed at Its End

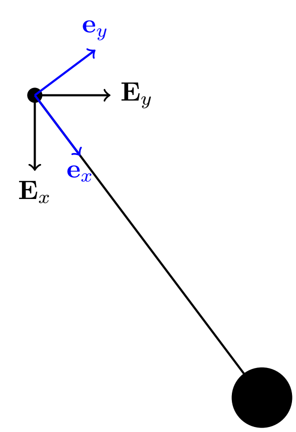


A bar of length ℓ and mass m pinned at its end. The energy is conserved:

$$U = -\frac{mg\ell}{2}\cos\theta, \tag{23.14}$$

$$E = \frac{m\ell^2}{6}\dot{\theta}^2 - \frac{mg\ell}{2}\cos\theta = \text{const.} \tag{23.15}$$

23.4.2 Bar Pendulum With End Mass



Adding a dead mass m_e at the end:

$$U = -\frac{mg\ell}{2}\cos\theta - m_e g\ell\cos\theta. \tag{23.16}$$

23.4.3 Follower Force

If instead a follower force $\mathbf{P}$ is applied perpendicular to the bar at its end, this force is **nonconservative** (its direction actively changes with configuration). Energy is not conserved.

23.5 Energy Conservation

As with particles and systems of particles, this theorem can be used to establish conservation of the total mechanical energy of a rigid body, $E = T + U$, when all work-doing forces and moments are conservative.

24 Summary

In summary, this book has no content whatsoever.

References

Lubliner, Jacob, and Panayiotis Papadopoulos. 2017. *Introduction to Solid Mechanics: An Integrated Approach*. Springer. https://doi.org/10.1007/978-1-4614-6768-7.

Part V

Appendices

25 Mathematical Background for Conservative Forces

25.1 Gradient

In coordinate-free terms, the gradient of a function $f(\mathbf{r})$ is defined by

$$df = \nabla f \cdot d\mathbf{r}, \tag{25.1}$$

where df is the total infinitesimal change in f for an infinitesimal displacement $d\mathbf{r}$.

25.1.1 Cartesian Coordinates

For $f = f(x, y, z)$ with $d\mathbf{r} = dx\mathbf{E}_x + dy\mathbf{E}_y + dz\mathbf{E}_z$:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz. \tag{25.2}$$

Since this must hold for all independent variations dx, dy, dz:

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{E}_x + \frac{\partial f}{\partial y}\mathbf{E}_y + \frac{\partial f}{\partial z}\mathbf{E}_z. \tag{25.3}$$

25.1.2 Cylindrical-Polar Coordinates

For $f = f(r, \theta, z)$ with $d\mathbf{r} = dr\mathbf{e}_r + r\,d\theta\mathbf{e}_\theta + dz\mathbf{E}_z$:

$$df = \frac{\partial f}{\partial r}dr + \frac{\partial f}{\partial \theta}d\theta + \frac{\partial f}{\partial z}dz = \nabla f \cdot (dr\mathbf{e}_r + r\,d\theta\mathbf{e}_\theta + dz\mathbf{E}_z). \tag{25.4}$$

Matching coefficients:

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{E}_z. \tag{25.5}$$

25.1.3 Curvilinear Coordinates

> **Note**
>
> This section is beyond the scope of this course. See O. M. O'Reilly's *Intermediate Dynamics* Section 1.6 for more information.

Consider a curvilinear coordinate system $\{q^1, q^2, q^3\}$ defined by $q^i = \hat{q}^i(x_1, x_2, x_3)$, assumed locally invertible. The **covariant basis vectors** are:

$$\mathbf{a}_i = \frac{\partial \mathbf{r}}{\partial q^i} = \sum_{k=1}^{3} \frac{\partial \hat{x}^k}{\partial q^i} \mathbf{E}_k. \tag{25.6}$$

The **contravariant basis vectors** are the gradients of the curvilinear coordinates:

$$\mathbf{a}^k = \nabla q^k = \sum_{i=1}^{3} \frac{\partial \hat{q}^k}{\partial x_i} \mathbf{E}_i. \tag{25.7}$$

25.2 Condition for a Vector Field to Be Conservative

A force $\mathbf{F} = F_x \mathbf{E}_x + F_y \mathbf{E}_y + F_z \mathbf{E}_z$ is derivable from a potential if $F_x = \partial f / \partial x$, $F_y = \partial f / \partial y$, $F_z = \partial f / \partial z$. This requires:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}, \qquad \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}, \qquad \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y}, \tag{25.8}$$

summarized by $\operatorname{curl} \mathbf{F} = \mathbf{0}$.

This condition is both necessary and sufficient for $\mathbf{F}$ to be derivable from a potential (see this proof).

25.3 Conservative Forces

A conservative force $\mathbf{F}_c$ is derivable from a potential U:

$$\mathbf{F}_c = -\nabla U. \tag{25.9}$$

The work of a conservative force depends only on the endpoints:

$$W_{\mathbf{F}_c, AB} = \int_{\mathbf{r}(t_A)}^{\mathbf{r}(t_B)} \mathbf{F}_c \cdot d\mathbf{r} = -(U_B - U_A) = U_A - U_B. \tag{25.10}$$

25.3.1 Constant Force

For a constant force $\mathbf{C}$, the potential is $U = -\mathbf{C} \cdot \mathbf{r}$. In Cartesians:

$$\nabla(\mathbf{C} \cdot \mathbf{r}) = \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial x}\mathbf{E}_x + \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial y}\mathbf{E}_y + \frac{\partial \mathbf{C} \cdot \mathbf{r}}{\partial z}\mathbf{E}_z = \mathbf{C}. \tag{25.11}$$

Hence $\mathbf{C} = -\nabla(-\mathbf{C} \cdot \mathbf{r})$.

25.3.2 Spring Force

For the spring force $\mathbf{F}_s = -K(r - \ell_0)\mathbf{e}_r$ (taking the anchor at the origin), the potential is $U = \frac{1}{2}K(r - \ell_0)^2$. Using the polar gradient:

$$\nabla\left(\frac{1}{2}K(r - \ell_0)^2\right) = K(r - \ell_0)\mathbf{e}_r, \tag{25.12}$$

so $\mathbf{F}_s = -\nabla U$.

25.3.3 Gravitational Force

For Newton's law of gravitation, the force on particle 1 due to particle 2 is

$$\mathbf{F}_1 = G\frac{m_1 m_2}{\left\|\mathbf{r}_{2/1}\right\|^3}\mathbf{r}_{2/1}. \tag{25.13}$$

Using the polar gradient with $\mathbf{r}_{2/1} = r\mathbf{e}_r$, one can show that $U = -Gm_1 m_2/r$ is the corresponding potential.

26 Sets and Linear Function Spaces

Note

This appendix is taken from Professor Panayiotis Papadopoulos's notes for UC Berkeley's ME280A course, Section 2.1.

26.1 Sets

A *set* $\mathcal{U}$ is a collection of elements. Elements of the set are denoted $u \in \mathcal{U}$. A set can be finite or infinite, and may have algebraic structure imposed on it.

26.2 Linear (Vector) Spaces

A set $\mathcal{V}$ is a **linear space** (or vector space) over the real numbers $\mathbb{R}$ if it is equipped with two operations:

1. **Addition:** $\mathbf{u} + \mathbf{v} \in \mathcal{V}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$,
2. **Scalar multiplication:** $\alpha\mathbf{u} \in \mathcal{V}$ for all $\alpha \in \mathbb{R}$, $\mathbf{u} \in \mathcal{V}$,

satisfying the standard axioms (commutativity, associativity, existence of zero element, existence of additive inverse, distributivity).

Examples of linear spaces: $\mathbb{R}^n$; the space of continuous functions on an interval; the space of $n \times n$ matrices.

26.3 Linear Transformations

A map $\mathbf{T} : \mathcal{V} \to \mathcal{W}$ between two linear spaces is a **linear transformation** if:

$$\mathbf{T}(\mathbf{u} + \mathbf{v}) = \mathbf{T}\mathbf{u} + \mathbf{T}\mathbf{v} \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathcal{V}, \tag{26.1}$$

$$\mathbf{T}(\alpha\mathbf{u}) = \alpha\mathbf{T}\mathbf{u} \quad \text{for all } \alpha \in \mathbb{R},\ \mathbf{u} \in \mathcal{V}. \tag{26.2}$$

When $\mathcal{V} = \mathcal{W}$ is the space of 3D vectors $\mathbb{E}^3$, a linear transformation from $\mathbb{E}^3$ to $\mathbb{E}^3$ is called a **tensor**.

> **!** Note!
>
> The rotation $\mathbf{Q}$ is a linear transformation from $\mathbb{E}^3$ to $\mathbb{E}^3$, so it is a tensor. This justifies calling it the **rotation tensor** throughout the kinematics of rigid bodies chapter.

26.4 Basis and Dimension

A set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a **basis** for $\mathcal{V}$ if: - The vectors are **linearly independent**: $\sum_i \alpha_i \mathbf{e}_i = \mathbf{0}$ implies $\alpha_i = 0$ for all i. - They **span** $\mathcal{V}$: every $\mathbf{v} \in \mathcal{V}$ can be written as $\mathbf{v} = \sum_i v_i \mathbf{e}_i$.

The number of basis vectors is the **dimension** of the space.

For 3D Euclidean space $\mathbb{E}^3$, any orthonormal set $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ with $\mathbf{E}_i \cdot \mathbf{E}_j = \delta_{ij}$ is a basis.